Software Requirements Specification for Squigglr: A Browser-Based Visual-Motor Practice and Progress-Tracking Platform

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Revision History

Date	Developer(s)	Change
13 October 2026	Rayan Nasrallah	Added the de-identified specialist interview transcript as an appendix (#84).
14 October 2026	Rayan Nasrallah	Wrote Section 1, Purpose of the Project, with the six goals and their constants (#85).
14 October 2026	Rayan Nasrallah	Wrote Section 2, Stakeholders, including personas (#86).
14 October 2026	Rayan Nasrallah	Wrote Section 3, Mandated Constraints (#87).
14 October 2026	Rayan Nasrallah	Wrote Appendix A, Requirement Sources (#89).
14 October 2026	Rayan Nasrallah	Linked interview citations to transcript turns (#95).
14 October 2026	Rayan Nasrallah	Wrote Section 5, Relevant Facts and Assumptions, with business rules (#88).
14 October 2026	Rayan Nasrallah	Made child-dependent thresholds adult-configurable, added the improvement rule and SRC-008 (#102).
14 October 2026	Rayan Nasrallah	Wrote Section 18, Open Issues, and Section 27, Likely and Unlikely Changes (#90).
15 October 2026	Rayan Nasrallah	Added the specialist questionnaire responses as SRC-009, with an appendix.
15 October 2026	Ahmed Elzaria	Completed SRS front matter and conventions, introduced the glossary and constants register, and prepared the shared requirement structure (#131).

Date	Developer(s)	Change
14 October 2026	Rayan Nasrallah	Added safety-critical requirements SAF-RAY-001 to SAF-RAY-008 to Section 12.2 (#91).
14 October 2026	Moustafa Moustafa	Wrote Section 8.2, Product Use Case Table (#108).
15 October 2026	Rayan Nasrallah	Added my reflection responses to the appendix and my generative AI use disclosure (#139).
15 October 2026	Ahmed Elzaria	Specified guided learning activities and formal lesson behaviour; clarified pass-based advancement within adult-selected lessons (#132).
15 October 2026	Ahmed Elzaria	Added safety-critical requirements SAF-AHM-001 and SAF-AHM-002 for drawing-input integrity (#135).
15 October 2026	Moustafa Moustafa	Let the revision history table continue across pages (#108).
15 October 2026	Moustafa Moustafa	Wrote Section 6, The Scope of the Work, with the context diagram (#109).

Introduction

Purpose and intended readers. This Software Requirements Specification (SRS) defines the externally observable behaviour, constraints and quality requirements of Squigglr. It establishes a shared basis for the development team, the supervisor, reviewers and future maintainers to understand, design, implement and verify the product. It specifies *what* Squigglr must provide rather than prescribing an implementation, except where a stakeholder has imposed a constraint. The project serves children practising drawing on a stylus-enabled touchscreen and the supporting adults who configure activities and review practice records (SRC-001, SRC-004). The scope, stakeholders, goals and constraints are established in Sections 1–3; requirements are refined in Section 9 and the subsequent quality sections.

Template and adaptations. This document follows the Volere Requirements Specification Template by Suzanne Robertson and James Robertson [1]; its LaTeX skeleton was adapted from the course's Volere template. The team retains the original headings except where changes clarify the capstone scope. In particular, Section 4 includes a new symbolic-constants register; Section 9 separates learning activities, formal specification, creative drawing, attempt records and progress summaries; and Section 27 records likely and unlikely changes. Section 28 provides cross-document traceability, while Appendix 29 records requirement evidence. A separate Hazard Analysis covers risks, failures and mitigation measures rather than repeating Volere's risks material. The source register includes anonymized interview paraphrases with speaker codes and timestamps, and questionnaire statements identified by respondent and question code. The de-identified interview transcript is reproduced in Appendix 30, providing direct links to the supporting interview turns (SRC-005).

Conventions and requirement format. Project goals use `G.1`–`G.6`; constraints use `CON-###`; facts `FACT-###`; business rules `BR-###`; assumptions `ASM-###`; and fictional personas `PER-#`. Requirement sources use `SRC-###`; numbered statements within a source use `SRC-###.##`; and interview participants use anonymized codes `P1`, `P2` and `P3`. Product use cases use `UC-###`; open questions use `OI-##`; and likely and unlikely changes use `LC-###` and `ULC-###`, respectively. Functional requirements have the prefixes `FR-LRN`,

`FR-CRT`, `FR-REC` and `FR-PRG`, each followed by a three-digit suffix. Non-functional requirements use their section's assigned prefix. Hazard Analysis uses owner-prefixed identifiers such as `H-AHM`, likewise followed by a three-digit suffix. Identifiers, once assigned, are not renumbered. Each requirement contains a short title, one testable "shall" statement, rationale where needed, a fit criterion, a priority, a phase, evidence sources and links to related use cases, requirements or hazards. *Must*, *Should* and *Could* identify delivery priority. Delivery phases are *PH-1* (PoC demo: Nov. 23, 2026), *PH-2* (Rev 0: Feb. 1, 2027), *PH-3* (final demo: Mar. 15, 2027), and *Future*. These phase identifiers are distinct from the interview participant codes; the requirement-by-phase plan belongs in Section 21. Symbolic constants are written in monospaced capitals (for example, `STAGE_PASS`) and defined once in Section 4.2. A value marked *TBD* is not an approved threshold; unresolved values appear in the constants register, and are also linked to an open issue in Section 18 where such an issue exists. Diagrams, if used, must include a legend identifying their symbols and arrow meanings. Section 9.2 defines the state-machine notation used to specify guided-lesson behaviour.

Document roadmap. Sections 1–5 establish the problem, stakeholders, constraints, terminology and assumptions. Sections 6–9 describe the work and product boundaries, data and use cases, and functional behaviour. Sections 10–17 specify quality and compliance requirements. Sections 18–26 document open questions, reusable solutions, project implications, implementation phases and deferred features. Sections 27–28 cover change expectations and traceability. The appendices contain the requirement sources register, de-identified specialist interview transcript, follow-up specialist questionnaire (SRC-009), generative AI-use disclosure and team reflection. Each section range links to its first section; the table of contents provides links to individual sections.

1 Purpose of the Project

This section states the problem Squigglr addresses, who has it, and the goals the project commits to. The problem and goals come from the Problem Statement and Goals (PSG) document (SRC-001).

1.1 User Business

Visual-motor integration is the ability to coordinate what a person sees with how they move. Children rely on it when drawing, copying shapes and learning to write (SRC-001). Kindergarten and primary school learners need chances to practise these skills at a suitable level and to complete activities with less help over time. Older children who need continued support, including some children with autism, attention-deficit/hyperactivity disorder (ADHD) or other developmental or learning differences, share this need. A diagnosis alone does not determine the support a child requires (SRC-001).

The adults around these children have a related problem. Parents, guardians, caregivers and educators need to understand which difficulties a child met and how much guidance the child needed. A finished drawing alone does not show whether the child completed it independently, relied on tracing guides, or corrected mistakes during the attempt (SRC-001). Squigglr is a website, used on a touchscreen device with a stylus, that offers guided drawing practice, creative drawing, and records that help adults understand that practice. It makes no diagnosis or treatment claims (SRC-001).

The gap left by current alternatives, as the PSG frames it, is that a drawing's final result is kept without the process behind it: the guidance used, the difficulties met and the corrections made. Squigglr's value is in keeping that process visible to supporting adults while children practise with gradually less support. The comparison with existing products appears in the Off-the-Shelf Solutions section of this document.

1.2 Goals of the Project

Table 2 lists the six project goals, G.1 to G.6, using the PSG's goal titles (SRC-001). Each goal is stated as Purpose, Advantage and Measurement: what the goal is, the benefit it brings and to whom, and how the team will check that it is met. Following the teaching assistant's (TA's) request for measurable goals (SRC-003) and the supervisor's direction that no requirements are gathered from children (SRC-004), every check is done by the team or by adult reviewers using prepared or adult-made attempts, never with child participants. The acceptance parameters are named in Section 4.2. On the supervisor's advice (SRC-008), parameters that depend on the child are selected by a supporting adult within specified ranges; tests check that the selected settings and defaults are respected. Each lesson (for example,

drawing a square) moves through fixed stages that give gradually less help: a demonstration, tracing a full outline, drawing between endpoints only, and drawing with no guide. This sequence follows the activity the team presented to the domain specialists (SRC-005, Moustafa, 6:03).

G.3 was revised after the PSG was submitted. The submitted PSG said Squigglr adjusts guidance and difficulty automatically. Following the domain-specialist interview (SRC-005, P1, 28:37; P2, 29:18) and the Oct. 7 TA discussion (SRC-006), the team assigned *lesson selection, starting-stage selection and discretionary stage changes* to a supporting adult (G.3). A passing drawing attempt still advances automatically to the next stage *within the lesson the adult selected* (G.2, BR-002); completing a lesson never automatically selects another lesson.

Table 2: Project goals

Goal	**Purpose, advantage and measurement**
G.1 Support structured drawing practice	*Purpose:* every lesson in the agreed catalogue covers lines, shapes or simple drawings, starts with a demonstration, and gives feedback and a retry at each drawing stage (SRC-001). *Advantage:* structured practice at a suitable level supports visual-motor development (SRC-001), and once an adult starts a lesson the child practises without constant adult help (SRC-005, P1, 34:13). Within a lesson the child does not navigate; Squigglr moves between stages. Serves children. *Measurement:* met when the team completes every stage of every catalogue lesson with prepared correct attempts, each attempt passes, and each drawing stage accepts a drawing, gives feedback and allows a retry.

Goal	Purpose, advantage and measurement
G.2 Support progression towards independent drawing	*Purpose:* within an adult-selected lesson, Squigglr automatically advances to the next drawing stage when an attempt meets that stage's pass criteria (`STAGE_PASS`, chosen by the adult within a team-set range). Passing the final independent stage completes the current lesson; it never starts a different lesson. An attempt that does not meet the criteria is not passed, and the child can retry. This checks the drawing against fixed criteria and does not assess the child (SRC-001). *Advantage:* children need to perform activities with gradually less help (SRC-001); a child practises one level until proficient before moving on (SRC-005, P1, 28:37), and children reach these skills at different points (SRC-005, P1, 20:27). Serves children and supporting adults. *Measurement:* met when (a) prepared correct attempts advance from outline tracing to endpoint guidance and from endpoint guidance to independent drawing, and a correct independent attempt completes only the current lesson; (b) every prepared incorrect attempt, such as one with a missing side or a wrong endpoint, is not passed and its predefined difficulty is reported; and (c) every saved attempt records the stage it was made at, with no mislabelled record; and (d) for prepared attempts scored just above and just below the chosen `STAGE_PASS`, at its minimum, default and maximum, each attempt passes or fails as expected.

Goal	Purpose, advantage and measurement
G.3 Let supporting adults control support and difficulty	*Purpose:* adults choose which lesson a child practises and may let the child start at a later stage. Changing to another lesson, skipping a stage or moving backward requires an adult's authorization; routine forward advancement on a passing attempt within the chosen lesson does not (G.2). Squigglr may suggest returning to an earlier stage after `DIFFICULTY_STREAK` failed attempts in a row; the change applies only when an adult accepts it, and a declined or ignored suggestion changes nothing. To avoid endless repetition, the lesson ends after `ATTEMPT_LIMIT` (larger than `DIFFICULTY_STREAK`) unsuccessful attempts at one stage, and the furthest stage reached is saved. If the child's attempts at a stage are improving, Squigglr holds back the suggestion and allows `ATTEMPT_EXTENSION` more attempts, up to `ATTEMPT_LIMIT_MAX` in total, so that the limit does not cut off a child who is getting closer to passing. Improvement compares the child's own recent attempts with each other against the stage's fixed criteria; it is not shown as a score and does not assess the child. The adult chooses these values for each child within a team-set range (SRC-008). *Advantage:* automatically choosing a new lesson or changing the learner's starting level without adult direction would require an out-of-scope readiness assessment; the specialists advised adult control of such decisions (SRC-005, P1, 28:37; P2, 29:18; SRC-006). Serves children and supporting adults. *Measurement:* met when, for prepared sequences of successful, unsuccessful and mixed attempts, every suggestion shown matches the agreed criteria, each lesson without improvement ends exactly at `ATTEMPT_LIMIT` and each improving one is extended by `ATTEMPT_EXTENSION` and never passes `ATTEMPT_LIMIT_MAX`, with its progress saved, no lesson or starting stage changes without an adult's action, and these hold at the minimum, default and maximum of each value.

Goal	Purpose, advantage and measurement
G.4 Support editable creative drawing and refinement	*Purpose:* in the creative space, Squigglr refines a child's drawing of a shape or object in the agreed supported set by adjusting the child's own strokes; the result stays editable and the original can be restored (SRC-001; Moustafa, 6:03). *Advantage:* creative drawing gives room for self-expression, and editability keeps the child in control of the work (SRC-001). Serves children. *Measurement:* using prepared drawings of each item in the supported set, met when for every drawing (a) every point of the refined drawing lies within `REFINE_TOLERANCE` (TBD) of the original strokes; (b) the refined parts can be moved, resized, recoloured, erased and undone; (c) restoring returns the original strokes exactly; and (d) a failed refinement request leaves the drawing unchanged.
G.5 Preserve meaningful activity records	*Purpose:* each attempt's record keeps the stage, every predefined difficulty observed, later corrections and the final outcome; each lesson's record keeps the furthest stage reached and the successful attempts at each stage (SRC-001). *Advantage:* a finished drawing alone hides how much guidance was used and the corrections made (SRC-001). Serves supporting adults. *Measurement:* met when, for prepared attempts and lessons with known actions, stages and outcomes, every saved record matches its known contents, with no difficulty lost after a correction.

Goal	Purpose, advantage and measurement
G.6 Make progress understandable to supporting adults	*Purpose:* summaries show a child's progress over time, separate attempts made without a guide from attempts made with one, and link to the attempts behind them (SRC-001). *Advantage:* adults need to see changes in independent drawing to decide on further practice (SRC-001); the specialists suggested giving the adult a view of the child's progress alongside control of activities (SRC-005, P2, 29:18). Serves parents, guardians, caregivers and educators. *Measurement:* met when, for prepared histories showing improvement, recurring difficulties and changing stages, every summary separates the two kinds of attempt correctly and links to the right attempts; and at least `ADULT_REVIEWERS` (TBD) adult reviewers interpret example summaries and locate the supporting attempts, with at least `REVIEW_PASS_RATE` (TBD) of their tasks succeeding.

The symbolic constants used to evaluate these goals, including values configurable by supporting adults, are recorded in the central register (Table 10, Section 4.2). Unknown adult-configurable ranges, midpoint defaults and the improvement margin are defined in Section 4.2 (SRC-008).

2 Stakeholders

This section identifies who has an interest in Squigglr, who uses it, and how each group informed the requirements. Sources are cited by their Requirement Source (SRC) identifiers from Appendix A. The development team is not listed as a stakeholder.

Each stakeholder is classed by how directly they use Squigglr. *Primary* stakeholders use it directly: children, parents, guardians and caregivers, and educators and learning-support staff. *Secondary* stakeholders shape it or are affected by it without using it: the client, school administrators, and the domain specialists. *Tertiary* stakeholders have an indirect interest: occupational therapists and educational specialists, and the TA and course instructors.

2.1 Client

The client, a secondary stakeholder, is Dr. Christopher Anand, the project supervisor, who acts in a client-like role (SRC-003). He provides guidance on scope, feasibility, and technical direction (SRC-001). His binding decisions are recorded in the Mandated Constraints section (SRC-004).

2.2 Customer

Squigglr has no paying customer; it is a capstone project and no sale or purchase is planned. The adults who would choose to adopt it act as customers in practice: parents, guardians, and caregivers for home use, and educators and school administrators for school use (SRC-001).

2.3 Other Stakeholders

Table 3 lists the stakeholders other than the client and the hands-on users, with the knowledge each provides and how much they are involved and can influence the project. Conflicts between stakeholders are handled in the Priorities Assigned to Users section.

Table 3: Other stakeholders: knowledge needed, involvement and influence

Stakeholder	**Interest and knowledge needed**	**Involvement and influence**
School administrators (secondary)	Suitability for the school setting, device availability, and the resources needed (SRC-001).	Not yet involved. May influence adoption if Squigglr is considered for school use.

Stakeholder	Interest and knowledge needed	Involvement and influence
Domain specialists P1, P2 and P3 (secondary)	Three specialists with doctorates in human-computer interaction and related engineering fields who build software for children with autism, attention-deficit/hyperactivity disorder (ADHD) and other learning needs. They provide expert knowledge of the intended users (SRC-005). They preferred not to be named and are de-identified (SRC-006).	Interviewed on October 1, 2026 (SRC-005). They offered to help guide a future pilot (SRC-005, P1, 24:03), which supports involving them in verification and validation (SRC-006). They advise; they do not decide.
Occupational therapists and educational specialists (tertiary)	Progress milestones, benchmarks and feedback wording, as the specialists recommended (SRC-005, P2, 31:42).	Not yet consulted.
TA and course instructors (tertiary)	Assess the project deliverables.	Review each deliverable; they influence documentation, not product behaviour.

2.4 Hands-On Users of the Project

Squigglr has two groups of hands-on users, both primary stakeholders: children, who complete drawing activities, and supporting adults, who choose activities and review records.

Children. Kindergarten and primary school learners, and older children who need continued support, including children with autism, ADHD, or other developmental or learning differences (SRC-001). The intended age range is not yet defined; it is recorded as an open item, `AGE_RANGE` (TBD). Their characteristics vary widely, and a diagnosis alone does not determine the support a child needs (SRC-001; Rayan, 2:04). The characteristics below matter most for the requirements, because they shape how children read,

draw and move through the application without help:

- **Reading ability:** some children cannot read yet, or may not understand written instructions. Squigglr has to communicate in ways that do not depend on reading, such as images and icons (SRC-005, P1, 34:13) and spoken audio that can be muted (SRC-005, P2, 37:39).
- **Motor skills:** fine-motor ability varies; some children find holding a stylus difficult, and grip can affect how they draw (SRC-005, P3, 25:36). Fine-motor control is part of visual-motor practice, so children need to practise holding a tool, moving from larger tools to smaller ones closer to what schools use (SRC-005, P1, 26:46). Squigglr therefore uses a stylus as its main input. Drawing with a finger is the easiest starting point (SRC-005, P1, 26:46), and finger input is a stretch goal. Some children may also have gross-motor or vision challenges (SRC-005, P2, 19:05).
- **Support needs:** once an adult starts a lesson, he should be able to practise without constant adult help (SRC-005, P1, 34:13). He does not need to navigate, exit or restart a lesson. Adults decide when a child moves to a new lesson or skips a stage (SRC-005, P1, 28:37).
- **Device experience:** many young children already know how to navigate touchscreen applications on tablets (SRC-005, P1, 34:13).

Supporting adults. Parents, guardians, caregivers, educators, and learning-support staff. They set up accounts, choose lessons and starting stages, and review attempt records and progress summaries on a tablet or computer (SRC-001; P1, 34:13). Their technical experience is assumed to be that of an everyday web user (assumption). They need understandable information without a single overall score, diagnostic labels, or comparisons with other children (SRC-005, Moustafa, 6:03).

Experience levels. Table 4 rates each user group's subject-matter and technological experience on a scale of novice, journeyman and master.

Table 4: Experience of each hands-on user group

User group	**Subject matter (drawing and visual-motor practice)**	**Technology**
Children	Novice: they are still learning to draw lines and shapes (SRC-001).	Novice to journeyman: many can already navigate tablet applications (SRC-005, P1, 34:13), but some need adult help.
Parents, guardians, caregivers	Novice: not assumed to have training in visual-motor development (assumption).	Journeyman: everyday web users (assumption).
Educators, learning-support staff	Journeyman: they support learners' skill development as part of their work (assumption).	Journeyman: they use school-provided web tools (assumption).

Needs, concerns and their effect on requirements. Table 5 records what each user group needs or worries about, and how this section expects the requirements to respond. The concerns come from the sources cited; none were gathered from children (SRC-004).

Table 5: User needs and concerns, with the requirement response

User	**Need or concern (source)**	**Expected requirement response**
Children	May not read; need to practise without constant adult help (SRC-005, P1, 34:13).	An adult starts the lesson; within it the child does not navigate, and Squigglr moves between stages.
Children	Grip and fine-motor ability vary (SRC-005, P3, 25:36).	Stylus is the main input; finger input is a stretch goal.

User	Need or concern (source)	Expected requirement response
Children (guided lessons)	Attempts must show what the child actually drew (SRC-005, Moustafa, 6:03).	Records keep the child's actual strokes; no corrected shape is counted as their work.
Children (creative space)	Get a cleaner version of their own drawing that is still theirs (SRC-005, Moustafa, 6:03).	Refinement adjusts the child's strokes rather than replacing them; the result stays editable and the original can be restored.
Children	Two children with the same diagnosis may need different support (SRC-005, Rayan, 2:04).	Adults choose the lesson and starting stage for each child; nothing is set by diagnosis.
Caregivers	Become annoyed if they must intervene constantly (SRC-005, P1, 34:13).	Children can practise without constant adult help.
Caregivers, educators	Want to decide when a child moves to a new lesson or needs a different starting level (SRC-005, P1, 28:37; P2, 29:18).	Adults choose the lesson and starting stage and approve skipping or returning to an earlier stage. Passing an attempt advances only within that lesson.
Caregivers, educators	Need to see how far the child got in each lesson, how many attempts succeeded at each stage, and which difficulties recur (SRC-001).	Records save the furthest stage reached and the successful attempts at each stage. To avoid endless repetition, a lesson ends after `ATTEMPT_LIMIT` unsuccessful attempts at one stage, and the progress is saved. No single overall score or labels.
Administrators	Installing applications on school devices is difficult (SRC-003, SRC-004).	Browser-based; no installation.

2.5 Personas

The three personas below are **fictional**. They do not describe any real child or adult. Personas are used because requirements cannot be gathered from children directly, and the supervisor recommended them for this case (SRC-004). There are at most three, and each is a short life story of a person. Details that come from a source cite it; all other details were invented by the team to make the persona concrete. No information was collected from children (SRC-004). Each persona describes one individual; none represents all children or adults in its group.

The personas describe people, not the product. They leave out design choices such as screen layout or feedback wording, so they stay valid if those choices change. Each persona has a fixed identifier (PER-1 to PER-3) so that scenarios, use cases, usability tests and later documents can refer to it. The three personas are linked: Adam is Sarah's son and one of Janice's learners, so one scenario can follow him between home and school. Invented details should be checked with caregivers and educators when the opportunity arises; if one changes, the scenarios that depend on it must be reviewed.

PER-1 (learner): Adam. Adam is 15 years old; older children who need continued support are in scope (SRC-001). He lives with his parents and no siblings. Adam is autistic and has ADHD. He loves Spider-Man, colours Spider-Man pictures at home, and tends to fixate on things he enjoys.

- **Communication:** he is verbal but speaks in short, simple sentences, and most of his responses are memorized. He can say how he feels or what he wants, such as being hungry or wanting to play, but cannot hold the conversations typical for his age. He understands spoken words and instructions well.
- **Reading and writing:** he reads and writes only with a parent's support. He can read a few short words he recognizes, such as "yes" and "no", but not longer words or sentences.
- **Learning:** he learns best by watching someone do a task first and then copying it, rather than being told what to do without an example.
- **Attention:** he loses focus easily. For skill-building tasks such as reading or drawing, a caregiver needs to watch over him throughout. When

he colours, which he enjoys, he can work alone for a while, but needs encouragement every now and then to stay focused and finish.

- **Frustrations:** he becomes agitated when a task is repeated over and over.
- **Attitude to technology and tools:** he knows how to use an iPad. He has no experience with a stylus, but uses crayons, pens, pencils, coloured pencils and markers.
- **Goals:** finish a picture of something he likes and show it to his family.
- **Context of use:** at home with his mother nearby, and at school in sessions with Janice.

PER-2 (caregiver): Sarah. Sarah is Adam's mother and works full time. She wants Adam to keep building the drawing and fine-motor skills that support writing. She is not a specialist and learns about Adam's needs from his teachers and therapists.

- **Attitude to technology:** an everyday web user who wants things to work without setup.
- **Goals:** let Adam practise without being called over constantly (SRC-005, P1, 34:13); choose what Adam practises and decide when he moves on (SRC-005, P1, 28:37).
- **Review habits:** looks through Adam's practice history on a laptop once a week to see which stages he reached and what keeps going wrong (SRC-001). Shares what she sees with Janice at parent meetings.
- **Concerns:** does not want Adam reduced to a single score, labelled or compared with other children (SRC-005, Moustafa, 6:03); wants Adam's data kept private.

PER-3 (educator): Janice. Janice is a teacher at a school for students with special needs who works with several learners each day, including Adam. She has a short slot with each learner and little time to set things up between them.

- **Environment:** the school's devices are managed centrally, so she cannot install applications (SRC-003, SRC-004).
- **Attitude to technology:** comfortable with the school's web tools, but has no administrator rights.
- **Goals:** set a different activity for each learner (SRC-005, P2, 29:18); compare a learner's records with what she observes in class (SRC-001).
- **Concerns:** switching between learners quickly, and making sure one learner's records are never shown to another.

2.6 Priorities Assigned to Users

Table 6 assigns priorities. Teachers and students are the main priority (SRC-003). Rationale: children are the users the activities serve; educators and caregivers choose activities and act on the records, so their needs come next. Administrators affect adoption but not daily use, and the specialists advise rather than use Squigglr (SRC-006). When needs conflict, a child's ability to practise independently is balanced with adult control of lesson selection and discretionary stage changes; passing a stage advances within the selected lesson (G.2; SRC-005, P1, 28:37).

Table 6: User priorities

Priority	Users
Key	Children (learners); educators and learning-support staff (SRC-003).
Key	Parents, guardians, and caregivers (SRC-001); ranked key because they choose activities and review records at home.
Secondary	School administrators; domain specialists (SRC-006).

2.7 User Participation

Children do not take part in requirements gathering; their needs are inferred from adults and from personas (SRC-004). The requirements were informed by:

- the supervisor meeting (SRC-004) and TA meetings (SRC-003, SRC-006);
- a recorded interview with domain specialists P1–P3 (SRC-005), held because children could not be asked directly (SRC-004) and the specialists build software for the intended users (SRC-006);
- peer review of the Problem Statement and Development Plan (SRC-007);
- written answers from two of the three specialists to a follow-up questionnaire (SRC-009).

Parents, caregivers, and educators have not yet been consulted (SRC-005, Rayan, 12:31); this is an open item. Planned later participation includes adult reviewers in usability sessions (SRC-001), possible feedback from adults with autism or ADHD (SRC-005, P2, 37:39), and the specialists' offered support for a pilot (SRC-005, P1, 24:03). Any study involving children requires ethics approval first (SRC-004).

2.8 Maintenance Users and Service Technicians

During the project, the development team maintains Squigglr, including activity content and the hosted website. No maintenance users or service technicians outside the team have been identified. Who maintains Squigglr after the project ends is an open item (TBD).

3 Mandated Constraints

This section lists constraints imposed on Squigglr from outside the team: by the supervisor, the teaching assistant (TA), the domain specialists interviewed (SRC-005), the course and the university. They are not design choices. Each constraint has an identifier of the form CON-###, a description, a rationale, the party that imposed it, a fit criterion that shows whether it is met, and its sources from the Requirement Sources register (SRC-###). The constraints limit what the team may build; they do not choose a technology, library or design. For example, CON-001 requires a website but does not name a framework. Table 7 summarizes them.

Table 7: Summary of mandated constraints

ID	Constraint	Source
CON-001	Squigglr shall be a website used in a web browser.	SRC-003, SRC-004
CON-002	Squigglr shall require no installation on the user's device.	SRC-003, SRC-004
CON-003	Squigglr shall work on touchscreen devices with stylus input.	SRC-001, SRC-003–SRC-005
CON-004	Deliverables shall meet the course deadlines in Table 8.	SRC-002
CON-005	The project shall gather no information from or about children for requirements or evaluation.	SRC-004
CON-006	The project shall run no study involving children.	SRC-004, SRC-005
CON-007	Squigglr shall not change lessons, skip guidance stages or move backward without adult authorization; pass-based forward movement within the selected lesson is allowed.	SRC-001, SRC-005, SRC-006

3.1 Solution Constraints

The supervisor, the TA and the domain specialists set the form of the solution.

CON-001 Squigglr shall be a website used in a web browser.
Rationale: Installing apps on school devices is difficult, and for Canadian schools a website is much easier to adopt than an app-store app.
Imposed by: supervisor and TA.
Fit criterion: Squigglr opens and runs from a web address in a browser.
Source: SRC-003, SRC-004.

CON-002 Squigglr shall require no installation on the user's device.
Rationale: Shared school devices are often managed by the school, so adults may not be able to install software. Opening a web address is enough to start.
Imposed by: supervisor and TA.
Fit criterion: a first-time user starts an activity without installing any

app, plug-in or browser extension.
Source: SRC-003, SRC-004.

CON-003 Squigglr shall work on touchscreen devices with stylus input.
Rationale: Squigglr targets visual-motor skills, which include fine motor control. Children build that control by practising with a held tool, moving from large tools to smaller ones like pencils (SRC-005, P1, 26:46). Grip varies between children and affects how they draw (SRC-005, P3, 25:36), so finger input is kept as a stretch goal (SRC-001).
Imposed by: domain specialists, within the scope agreed with the supervisor and TA.
Fit criterion: each activity can be completed with a stylus on a touchscreen device; the device and browser combinations are TBD.
Source: SRC-001, SRC-003, SRC-004, SRC-005.

CON-007 Squigglr shall not select another lesson, skip a guidance stage or move backward to an earlier stage unless a supporting adult authorizes that change. Normal forward movement to the next stage following a passing attempt *within the selected lesson* is not an unauthorized change (G.2, BR-002).
Rationale: Choosing what a child is ready to practise requires judgement by the adult who knows the child (SRC-005, P1, 28:37; P2, 29:18). This restriction applies to lesson selection and discretionary stage changes, not to the fixed pass-based steps already defined for an adult-selected lesson. Squigglr may suggest going back a stage, but an adult must accept (G.3).
Imposed by: domain specialists' advice on adult choice, clarified by the team's progression decision.
Fit criterion: test that passing a tracing or endpoint stage advances within the current lesson without an additional adult action; passing the final stage completes that lesson without starting another. Verify that switching lessons, skipping and returning to an earlier stage cannot occur without an adult choice or acceptance.
Source: SRC-001, SRC-005 (P1, 28:37; P2, 29:18), SRC-006; team decision documented in #132.

3.2 Implementation Environment of the Current System

Not applicable. There is no current system that Squigglr replaces or must run alongside; it is a new product.

3.3 Partner or Collaborative Applications

Not applicable. No stakeholder requires Squigglr to work with a specific partner application. Candidate integrations, if any, belong in the Off-the-Shelf Solutions section.

3.4 Off-the-Shelf Software

Not applicable. No stakeholder mandates any specific off-the-shelf software. Candidate libraries and services are discussed in the Off-the-Shelf Solutions section, not here.

3.5 Anticipated Workplace Environment

Squigglr will be used at home, in classrooms and in other places where children practise drawing, such as rehabilitation centres (SRC-005, P2, 21:25), usually with an adult nearby. School devices may be shared by several children (SRC-004). These places give context for later requirements; they are not a constraint, since the team will not test Squigglr in each of them.

3.6 Schedule Constraints

The course sets the project's deadlines. Table 8 lists the course deadlines relevant to this document, taken from the Development Plan (SRC-002) course timeline. Dates use the America/Toronto time zone. The Development Plan holds the full timeline.

CON-004 Deliverables shall meet the course deadlines in Table 8.

Rationale: The capstone course fixes these dates; the team cannot change them.

Imposed by: the capstone course.

Fit criterion: each deliverable is submitted by its date.

Source: SRC-002.

Table 8: Course deadlines (from the Development Plan)

Date	Deliverable
Oct. 19, 2026	SRS and Hazard Analysis
Oct. 22, 2026	SRS peer-review issues
Nov. 2, 2026	Verification and Validation (V&V) Plan
Nov. 16, 2026	Design Revision −1 and fall custom document
Nov. 23–Dec. 4, 2026	Proof of Concept (POC) demonstration window
Jan. 18, 2027	Design documentation
Feb. 1–11, 2027	Rev 0 presentation sessions
Mar. 8, 2027	V&V Report and winter custom document
Mar. 15–21, 2027	Final demonstration sessions
Mar. 31, 2027	Final documentation

3.7 Budget Constraints

TBD. The Development Plan states no budget for the project, and no stakeholder has set one. It notes only that service limits and possible costs of external services must be monitored. The budget remains an open issue to confirm with the TA.

3.8 Enterprise Constraints

The supervisor and the university set rules that protect children.

CON-005 The project shall gather no information from or about children for requirements or evaluation.
Rationale: The users are children, and the supervisor directed that no information about them be collected. Requirements come from adults and personas, and every check uses prepared or adult-made attempts. This constraint covers the project's work; how Squigglr itself protects the practice records it keeps is a privacy requirement, not this constraint.
Imposed by: supervisor.
Fit criterion: no requirement source, test or evaluation record contains data from or about a child.
Source: SRC-004.

CON-006 The project shall run no study involving children.

Rationale: Any study run through the university that involves children requires ethics board approval, which the project does not seek (SRC-005, P2, 22:56; P1, 33:00). Evaluation is done with adults only (CON-005).

Imposed by: supervisor and the university, as the domain specialists also advised.

Fit criterion: no study with child participants takes place during the project.

Source: SRC-004, SRC-005.

4 Naming Conventions and Terminology

4.1 Glossary and Acronyms

Table 9 defines the terms and abbreviations used in the goals, business rules, stakeholder evidence and requirements. Source identifiers link to Appendix 29.

Table 9: Project terminology and acronyms

Term or acronym	Definition	Context
Visual-motor integration	Coordination of visual information with movement, relevant to drawing and writing-readiness activities.	Section 1
Guided lesson	A structured activity in the agreed lesson catalogue in which drawing guidance is provided in successive stages.	G.1–G.3
Stage (guidance level)	A step within a lesson that provides a specified amount of help, from demonstration through independent drawing. A passing drawing attempt advances automatically to the next stage within the chosen lesson; adult authorization is needed to skip or return.	G.2–G.3
Attempt	One try at the drawing task of a lesson stage; its strokes, observed difficulties and outcome are represented in the practice record.	BR-002, BR-004

Term or acronym	Definition	Context
Attempt record	The record of the actual strokes, stage, predefined observations and outcome of an attempt; guided records are not replaced by refined shapes.	BR-004
Creative refinement	Automatic, minimal adjustment of the child's strokes in the creative space only. The drawing remains editable, and the original can be restored by the child or, if needed, by a supporting adult.	BR-005
Supporting adult	A parent, guardian, caregiver, educator or learning-support staff member who configures activities and reviews records, as permitted by the account rules.	Section 2
Fit criterion	An observable or measurable acceptance check used to verify whether a requirement or constraint is met.	Section 3
ADHD	Attention-deficit/hyperactivity disorder.	Section 1
AI	Artificial intelligence.	BR-007
ASM	Assumption identifier; `ASM-###`.	Section 5.3
BR	Business rule identifier; `BR-###`.	Section 5.2
FACT	Relevant fact identifier; `FACT-###`.	Section 5.1
BUC	Business use case.	Document roadmap
CON	Mandated constraint identifier; `CON-###`.	Section 3
FR	Functional requirement.	Section 9
LC	Likely change identifier; `LC-###`.	Section 27.1
HA	Hazard Analysis, a separate document identifying failures, losses, mitigation measures and related requirements.	Introduction
NFR	Nonfunctional requirement; a required quality or constraint on system performance or operation.	Section 12
OI	Open issue identifier; `OI-##`.	Section 18
PER	Fictional persona identifier; `PER-#`.	Section 2.5
P1–P3	Anonymized identifiers for the three domain specialists interviewed as part of requirements elicitation; not delivery phases.	Appendix A: Interview statements

Term or acronym	Definition	Context
POC / PoC	Proof of concept; the planned capstone feasibility demonstration, not a production release.	Schedule
PSG	Problem Statement and Goals project document.	Section 1
PUC	Product use case.	Section 8
SRS	Software Requirements Specification, this document.	Introduction
SRC	Requirement source identifier; `SRC-###`. Numbered source statements use `SRC-###.##`.	Appendix A
TA	Teaching assistant.	Section 1
TBD	To be determined; a value or decision not yet approved.	Section 4.2
UC	Product use case identifier; `UC-###`.	Section 8
ULC	Unlikely change identifier; `ULC-###`.	Section 27.2
V&V	Verification and validation.	Schedule
WCAG	Web Content Accessibility Guidelines; selected criteria may inform project accessibility requirements, without a claim of complete conformance (SRC-008).	OI-08

4.2 Symbolic Constants

Table 10 specifies the parameters used in guided lessons, creative refinement and progress-summary verification. The six adult-configurable parameters have an allowed range and a default value for each learner profile (BR-008, SRC-008). Their defaults are the midpoints of their respective ranges. Each configuration shall satisfy `ATTEMPT_LIMIT` $>$ `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT_MAX` $\geq$ `ATTEMPT_LIMIT`. Drawing-stage results are expressed on a normalized scale from 0 to 100; higher results indicate closer agreement with the stage's fixed criteria. The result is not a measure of the child's ability (BR-006). Attempt-count settings shall be integers, and values violating the allowed ranges or the two cross-setting inequalities shall be rejected (BR-008).

Basis and verification of numerical values. The project team selected the initial ranges, defaults, five-point `IMPROVE_MARGIN` and 0–100 result scale in a team discussion on 15 October 2026 concerning issue #132. These are team-selected engineering starting values, not thresholds established by child testing or supplied numerically by the supervisor. The supervisor's 14 October guidance (SRC-008) requires adult-configurable ranges, midpoint defaults and example drawings illustrating both ends of the `STAGE_PASS` range. The chosen values and the drawing-to-score interpretation are checked using prepared or adult-made examples at the lower and upper boundaries of each configurable range (OI-05; CON-005–006).

Table 10: Central symbolic constants register

Constant	**Meaning, unit, value and traceability**
`STAGE_PASS`	Minimum stage-result threshold for a complete attempt to pass a drawing stage. *Unit:* points out of 100. *Range:* 70–90. *Default:* 80. *Trace:* G.2, BR-002, BR-008, OI-05.
`DIFFICULTY_STREAK`	Consecutive unsuccessful attempts before Squigglr may suggest returning to an earlier stage. *Unit:* failed attempts. *Range:* 3–5. *Default:* 4. *Trace:* G.3, BR-003, OI-05.
`ATTEMPT_LIMIT`	Number of unsuccessful attempts that triggers the lesson-end or extension decision. *Unit:* failed attempts. *Range:* 6–10. *Default:* 8; must exceed `DIFFICULTY_STREAK`. *Trace:* G.3, BR-002, OI-05.
`IMPROVE_WINDOW`	Number of evaluable attempts in each comparison window used by the improvement rule. *Unit:* attempts. *Range:* 1–3. *Default:* 2. *Trace:* G.3, BR-002, OI-05.
`IMPROVE_MARGIN`	Minimum increase in average stage-result points across consecutive comparison windows that qualifies as improvement. *Unit:* points out of 100. *Value:* 5 points (fixed by the team). *Trace:* G.3, BR-002, OI-05.

Constant	Meaning, unit, value and traceability
ATTEMPT_EXTENSION	Additional evaluable attempts permitted when recent results satisfy the improvement criterion. *Unit:* attempts. *Range:* 1–3. *Default:* 2. *Trace:* G.3, BR-002, OI-05.
ATTEMPT_LIMIT_MAX	Maximum number of evaluable attempts at a stage, including extensions. *Unit:* attempts. *Range:* 8–12. *Default:* 10; must be at least ATTEMPT_LIMIT. *Trace:* G.3, BR-002, OI-05.
REFINE_TOLERANCE	Maximum permitted distance between refined strokes and the child's original strokes in the creative space. *Unit:* distance scale (TBD). *Value:* team-defined; TBD. *Trace:* G.4, BR-005.
ADULT_REVIEWERS	Minimum number of adult reviewers for progress-summary understandability checks. *Unit:* people (count). *Value:* team-defined; TBD. *Trace:* G.6.
REVIEW_PASS_RATE	Minimum proportion of progress-summary review tasks completed successfully by adults. *Unit:* proportion. *Value:* team-defined; TBD. *Trace:* G.6.
AGE_RANGE	Intended lower and upper ages of learners served by Squigglr. *Unit:* years. *Value:* lower and upper bounds TBD. *Trace:* Section 2, G.1.

5 Relevant Facts And Assumptions

This section records what is known to be true now, the rules that govern how Squigglr may be used, and what the team believes but has not verified. Facts and assumptions are kept apart so that a reader can see which requirements rest on unconfirmed beliefs. The Hazard Analysis uses the assumptions as possible causes of hazards: if an assumption turns out to be false, the requirements that depend on it must be reviewed. Each entry identifies the goals, constraints or product behaviours it supports; Section 9 specifies the corresponding functional requirements.

5.1 Relevant Facts

Each fact below is true at the time of writing and is supported by a source in Appendix 29. Facts from the interview are expert statements, not observations of children.

FACT-001

Visual-motor integration is linked to handwriting legibility in kindergarten children, but the link is a correlation; it does not show that practising one improves the other (SRC-001; Rayan, 2:04).
Relevance: Squigglr makes no claim that its practice improves handwriting or visual-motor skills (BR-006).

FACT-002

The study of structured, graded practice cited in the Problem Statement had no control group. It supports the idea of graded practice but does not show that Squigglr will produce the same results (SRC-001; Rayan, 2:04).
Relevance: effectiveness is not a goal or a fit criterion of this project.

FACT-003

Motor and handwriting difficulties occur in some autistic children and some children with ADHD, with severity ranging from mild to severe; not every child in these groups has them (SRC-001; Rayan, 2:04).
Relevance: guidance and difficulty are set per child by an adult, not per diagnosis (G.3).

FACT-004

The specialists stated that deciding automatically when a child is ready for the next level requires assessment, which is too complex for the project's time frame, and that an adult who knows the child should decide (SRC-005, P1, 28:37; P2, 29:18).
Relevance: basis for adult-controlled lesson selection and nonroutine stage changes (CON-007, G.3, BR-003); the team retains fixed pass-based forward progression within the selected lesson (G.2).

FACT-005

The specialists stated that how a child holds a tool affects how they draw, that finger input is the easiest start, and that fine-motor practice needs a held tool, moving from large to small (SRC-005, P3, 25:36; P1,

26:46).
Relevance: basis for stylus input (CON-003) with finger input as a stretch goal.

FACT-006
The specialists stated that tracing in the air in front of a camera practises gross rather than fine motor skills, and that it should come after the touchscreen activities when time is limited (SRC-005, P1, 13:42; P1, 16:13; P2, 16:45).
Relevance: camera finger tracing stays a stretch goal.

FACT-007
The specialists stated that young children may not read yet but navigate games through images and icons, and that a child should be able to enter a lesson, practise, restart and move back and forth without constant adult help, while adults handle setup (SRC-005, P1, 34:13).
Relevance: G.1 and the usability requirements; requirement IDs to be linked when written.

FACT-008
A study that involves children requires ethics approval; meeting adult experts for feedback, without involving children, does not (SRC-004; SRC-005, P2, 22:56; P1, 33:00).
Relevance: basis for CON-005 and CON-006.

FACT-009
Installing applications on school-managed devices is difficult, and a website is easier for Canadian schools to adopt (SRC-003, SRC-004).
Relevance: basis for CON-001 and CON-002.

FACT-010
The specialists recommended starting from the Web Content Accessibility Guidelines (WCAG), which one participant stated are adopted in law in many countries, and then applying design heuristics for autism from the literature (SRC-005, P2, 39:26).
Relevance: accessibility and standards requirements; requirement IDs to be linked when written.

FACT-011
The course fixes the project's deadlines (CON-004, SRC-002).

Relevance: scope decisions, such as keeping assessment and camera tracing out of the core product, follow from the time available.

FACT-012

External requirements evidence includes a 43-minute interview with three domain specialists and a follow-up questionnaire answered by two of them. No parent, educator, school administrator or occupational therapist has been consulted, and no child has taken part in any source (SRC-005, SRC-009; SRC-004).

Relevance: requirements based on what parents and educators need rest on assumptions (ASM-006) until those groups are consulted.

FACT-013

The interview did not cover several topics that affect the requirements: how Squigglr compares with existing drawing apps (one participant noted only that many exist (SRC-005, P2, 35:36)), differences between home and school use, the relative priority of the features, and how practice records should be kept, shared or deleted (SRC-005).

The follow-up questionnaire later covered refinement in the creative space and letter activities (SRC-009).

Relevance: the requirements on these topics rest on assumptions (ASM-010 to ASM-012) and are open to change.

5.2 Business Rules

The rules below govern who may do what with Squigglr and what Squigglr must never do. They hold no matter how Squigglr is designed. Each rule has an identifier of the form BR-###, a rationale and its sources.

BR-001

Only the adult who holds an account, a parent or an educator, may create, view and manage the learner profiles under it and their records. One account can hold several learner profiles. A learner profile holds only the child's first name and a username that does not identify the child; no other information about the child is collected.

Rationale: practice records describe a child's difficulties and are personal information. Keeping each profile to a first name and a non-identifying username lets an adult tell children apart, for example on a shared device, while collecting as little about them as possible. How

records are protected is a privacy requirement.
Source: SRC-001, SRC-004; G.5, G.6.

BR-002

Within an adult-selected lesson, Squigglr automatically moves the child to the next guidance stage when an attempt meets the current drawing stage's pass criteria (`STAGE_PASS`). Passing the independent drawing stage completes the lesson; it does not select or start another lesson. After `ATTEMPT_LIMIT` (larger than `DIFFICULTY_STREAK`) unsuccessful attempts at one stage, the lesson ends, and the furthest stage reached and the number of successful attempts at each stage are saved. If, when `ATTEMPT_LIMIT` is reached, the average result of the last `IMPROVE_WINDOW` attempts is better than that of the `IMPROVE_WINDOW` attempts before them by at least `IMPROVE_MARGIN`, the child is improving and Squigglr grants a single extension of up to `ATTEMPT_EXTENSION` additional evaluable attempts. The extension cannot be granted a second time at the same stage, regardless of later result changes. Total evaluable attempts at a stage never exceed `ATTEMPT_LIMIT_MAX`.
Rationale: moving forward on a pass checks the drawing against fixed criteria and does not assess the child (G.2). Ending the lesson stops a child from repeating a stage they cannot pass without limit, and saving the progress lets the adult decide what comes next. Extending the limit while the child improves avoids cutting off practice that is working, and `ATTEMPT_LIMIT_MAX` keeps the stop. The comparison uses only the child's own attempts against the stage's fixed criteria, so it does not assess the child.
Source: G.2, G.3, G.5.

BR-003

A supporting adult chooses which lesson the child practises and its starting stage. Squigglr never changes lessons, skips a stage or moves back a stage on its own. Passing an attempt may advance the child to the next stage *within that same lesson* as defined by BR-002. After `DIFFICULTY_STREAK` failed attempts in a row, Squigglr may suggest going back a stage, unless the child is improving (BR-002); the change happens only if an adult accepts.
Rationale: deciding readiness needs assessment that is out of scope (FACT-004), and the adult knows the child's level.

Source: CON-007; G.3; SRC-005, P1, 28:37; P2, 29:18.

BR-004

A guided lesson record keeps the child's actual strokes. Squigglr never replaces them with a corrected or ideal shape.
Rationale: adults judge progress from the records; a record that shows a corrected shape would mislead them.
Source: G.5; SRC-005, Moustafa, 6:03.

BR-005

Refinement of strokes happens only in the creative space. It is automatic and minimal, the refined drawing stays editable, and the child's original drawing can be restored, by the child or, if the child cannot, by the adult.
Rationale: the work must remain the child's own, and refinement must never reach the records adults rely on (BR-004). Refinement needs no action from the child, and an adult can restore the original for a child who cannot use the control.
Source: G.4; SRC-001.

BR-006

Squigglr gives no diagnosis, no judgement of a child's abilities, no comparison with other children, and no claim that its practice improves any skill.
Rationale: the team are not occupational therapists and cannot judge a child's abilities (SRC-008), and the evidence does not support effectiveness claims (FACT-001, FACT-002).
Source: SRC-001, SRC-008; SRC-005, Moustafa, 6:03.

BR-007

Feedback during a guided activity comes from observations the team defines in advance, such as a missed part of an outline. It is not personalized advice generated by an AI model.
Rationale: predefined observations can be reviewed and checked before release; generated advice to a child cannot.
Source: SRC-001; SRC-005, Moustafa, 6:03.

BR-008

The supporting adult sets `STAGE_PASS`, `DIFFICULTY_STREAK`, `ATTEMPT_LIMIT`,

IMPROVE_WINDOW, ATTEMPT_EXTENSION and ATTEMPT_LIMIT_MAX for each child. When an adult creates a child profile, Squigglr asks them to choose each value, and for STAGE_PASS shows example drawings of what its minimum and maximum look like. The adult can change the values later. A value the adult does not choose is set to the middle of its range. Squigglr accepts only values within the range, only an ATTEMPT_LIMIT larger than DIFFICULTY_STREAK, and only an ATTEMPT_LIMIT_MAX at least ATTEMPT_LIMIT.
Rationale: the adult knows the child, and a value that suits one child may not suit another (FACT-003). The team sets only the minimum and maximum, so no single value has to be shown right for every child.
Source: SRC-008; CON-007; Table 10.

BR-009

The team sets the minimum and maximum of each adult-set value, and the fixed values in Table 10, using adult-made drawings, never from testing with children.
Rationale: no study with children takes place (CON-005, CON-006).
Source: SRC-004, SRC-008; Table 10.

5.3 Assumptions

The assumptions below are beliefs the team holds but has not verified. Each has an identifier of the form ASM-###, a rationale, what depends on it, and the effect if it is false. Assumptions marked as relevant to safety are inputs to the Hazard Analysis.

ASM-001

An adult is present when a child starts a guided lesson, to choose the lesson and the starting stage.
Rationale: the specialists described adults handling setup and deciding progression (SRC-005, P1, 28:37; P1, 34:13). How often this happens at home or in class is unknown.
Depends on it: G.3, BR-003, CON-007.
If false: a child may start at a level an adult did not choose. Relevant to safety.

ASM-002

A child may continue practising without an adult watching, but an

adult nearby remains responsible for the child's physical safety, wellbeing and time on the device. Squigglr cannot observe or prevent physical harm, distress or overuse.
Rationale: the specialists advised that children should practise without constant adult help (SRC-005, P1, 34:13), and the Problem Statement expects adult support as each learner needs (SRC-001).
Depends on it: G.1, the safety requirements (requirement IDs to be linked when written).
If false: the safety requirements must cover unsupervised use, for example limits on session length or sound. Relevant to safety.

ASM-003

Homes and schools that use Squigglr have a touchscreen device with a stylus that works in a supported browser.
Rationale: the stylus is the main input (CON-003), and the interview example used a tablet with a stylus (SRC-005, Rayan, 24:56). Which devices families and schools own is not known. The supported device and browser combinations are TBD.
Depends on it: CON-003, G.1, G.2.
If false: children without a stylus can still use Squigglr with a finger, but not as intended: the stage checks and value ranges are set for stylus input, and tuning them for finger input is a stretch goal.

ASM-004

Supported browsers report stylus strokes in enough detail (position and order of points) for Squigglr to check an attempt against `STAGE_PASS` and to store the actual strokes.
Rationale: the stage checks and the records both depend on the strokes; this has not been tested on any device yet.
Depends on it: G.2, G.5, BR-004.
If false: an attempt may be judged wrongly or stored inaccurately, which would mislead the adult. The team may then need to deliver Squigglr as an installed app, if input there is more reliable, which would change CON-001. Relevant to safety.

ASM-005

A network connection is available during normal use, for loading activities and for saving and viewing records. Offline use is out of the initial scope.

Rationale: stated in the Problem Statement (SRC-001); school and home networks have not been checked.
Depends on it: G.5, G.6, CON-001.
If false: attempts may be lost when the connection drops, and an adult may decide from incomplete records. Relevant to safety.

ASM-006

The supporting adult can judge from the records when a child should move on, stay or go back.
Rationale: the specialists placed this decision with a parent or educator who knows the child (SRC-005, P1, 28:37). Squigglr shows the adult every value it records, such as each attempt's result and the `STAGE_PASS` and `ATTEMPT_LIMIT` in use for that attempt, and the adult chooses which of them their view shows, so the team does not have to pick which measures matter.
Depends on it: G.3, G.6, CON-007, BR-003.
If false: a child may be kept at a level that is too hard or too easy; the adult views would need more guidance. Relevant to safety.

ASM-007

The adult who uses the adult views can read the language of the interface. The interface language is TBD. Children do not need to read to use the child views.
Rationale: adult tasks such as reviewing records need text, while child navigation uses images and icons (FACT-007).
Depends on it: G.6 and the usability requirements (requirement IDs to be linked when written).
If false: adults may misread records or settings.

ASM-008

On a shared device, an adult makes sure the right child is selected before a session and that another child cannot open the previous child's records.
Rationale: school devices may be shared by several children (SRC-004). One adult account can hold several learner profiles (BR-001).
Depends on it: BR-001, G.5.
If false: attempts may be saved to the wrong child, or a child's records may be seen by others. Relevant to safety and privacy.

ASM-009

The adults who use Squigglr for a child have the right to see that child's records, and any consent this needs is obtained outside Squigglr.
Rationale: Squigglr cannot verify a parent's or school's authority over a child; it only knows the adult who holds the account (BR-001).
Depends on it: BR-001, the privacy requirements (requirement IDs to be linked when written).
If false: an unauthorised adult could see a child's records. Relevant to privacy.

ASM-010

Lines, basic shapes and simple drawings are suitable practice content for the intended children, without letters.
Rationale: the specialists did not object to this content, and nurseries focus on these skills (SRC-005, P1, 20:27). Two specialists saw letters as a later stage after shapes (SRC-009.04), so they remain a stretch goal (SRC-001).
Depends on it: G.1, G.2.
If false: the activity set changes.

ASM-011

Children and adults value automatic refinement in the creative space, and children understand that the refined drawing is still theirs.
Rationale: questionnaire respondents supported subtle, optional refinement that preserves the child's drawing (SRC-009.01), but their responses do not establish children's or caregivers' preferences.
Depends on it: G.4, BR-005.
If false: refinement may discourage a child, and G.4 would need to change.

ASM-012

Squigglr serves the same needs at home and in school, so one set of requirements covers both.
Rationale: the specialists named homes, kindergartens, schools and rehabilitation centres as possible settings (SRC-005, P2, 21:25) but did not discuss differences between them (FACT-013).
Depends on it: G.3, G.6, BR-001.
If false: separate requirements may be needed, for example for an educator managing several children.

ASM-013

Schools allow access to the Squigglr website on their managed devices and networks.

Rationale: a website is easier to adopt than an installed app (FACT-009), but no school has confirmed that it would not block the site.

Depends on it: CON-001, CON-002.

If false: school use needs approval from administrators.

ASM-014

Strokes drawn by adults, or prepared examples, are close enough to children's drawing to verify stage checks and refinement.

Rationale: evaluation uses adults only (CON-005, CON-006).

Depends on it: G.2, G.4, BR-009, and the verification plan.

If false: the range of `STAGE_PASS` and the value of `REFINE_TOLERANCE` may be wrong for children, so attempts may be judged wrongly. Relevant to safety.

ASM-015

An adult can turn off audio feedback, or stop a session, if sound or on-screen feedback upsets a child.

Rationale: the specialists recommended that users can turn audio off (SRC-005, P2, 37:39). Children react to sensory feedback in different ways, and the team has no information on which feedback upsets which children.

Depends on it: the accessibility and safety requirements (requirement IDs to be linked when written).

If false: feedback must be safe by default rather than relying on an adult. Relevant to safety.

ASM-016

A supporting adult can choose adult-set values (Table 10) that suit their child, helped by the example drawings for `STAGE_PASS`, or leave the defaults.

Rationale: the supervisor advised leaving these values to the adult (SRC-008); no adult has tried the choice yet.

Depends on it: G.2, G.3, BR-008.

If false: a child may be held at a stage too long or moved on too soon, which can frustrate them. Relevant to safety.

ASM-017

A rise in the average result of a child's recent attempts by at least `IMPROVE_MARGIN` reflects real improvement, not chance variation between drawings.

Rationale: drawing results vary from one attempt to the next; the team will check the margin with prepared adult-made sequences, not with children (CON-005, CON-006).

Depends on it: G.3, BR-002.

If false: a lesson may be extended for a child who is not improving, or not extended for one who is. `ATTEMPT_LIMIT_MAX` still ends the lesson. Relevant to safety.

6 The Scope of the Work

This section describes the work that Squigglr will become part of: children practising drawing, and the adults around them choosing that practice and following the child's progress. It covers how the work is done now, the boundary of the work and what crosses it, the business events the work responds to, and the business use case for each event. Squigglr's own part of the work is in Section 8.

6.1 The Current Situation

Children practise drawing lines, shapes and simple drawings at home, in nurseries and kindergartens, in schools and in rehabilitation centres (SRC-005, P1, 20:27; P2, 21:25). They practise with tools they hold, starting with large ones that are easier for their hands and moving to smaller ones, such as pencils, that are closer to what they will use in school (SRC-005, P1, 26:46). Some children also take part in therapy programmes run by occupational therapists (SRC-005, P1, 32:42). Structured programmes exist that grade the same activity into easier and harder levels (SRC-001), and many drawing applications are available for children (SRC-005, P2, 35:36). The comparison with existing products is in the Off-the-Shelf Solutions section.

The adults who support a child follow progress mainly through what the child produces. A finished drawing does not show whether the child made it alone, relied on tracing guides, or corrected mistakes along the way (SRC-001). The adult who knows the child decides when the child is ready to move

on (FACT-004). The work as it is done now therefore has three gaps that the project goals address:

- practice is not always graded so that help fades as the child improves (G.1, G.2);
- the adult's decision on the next step rests on finished drawings, not on a record of how they were made (G.3, G.5);
- nothing summarizes the child's practice over time for the adult (G.6).

6.2 The Context of the Work

Figure 1 shows the boundary of the work and the adjacent actors that exchange information with it, and Table 11 lists every data flow that crosses the boundary. The work includes Squigglr and the decisions supporting adults make with it, such as which lesson a child practises. The child and the supporting adult are drawn outside the boundary because they start the work and receive its results. The Squigglr team is an adjacent actor because it maintains the lesson catalogue and sets the ranges of the adult-set values (BR-009). Children and adults reach the work through a touchscreen device with a stylus in a web browser (CON-001, CON-003). The device carries the flows; it is not an adjacent actor because it neither starts the work nor uses its results.

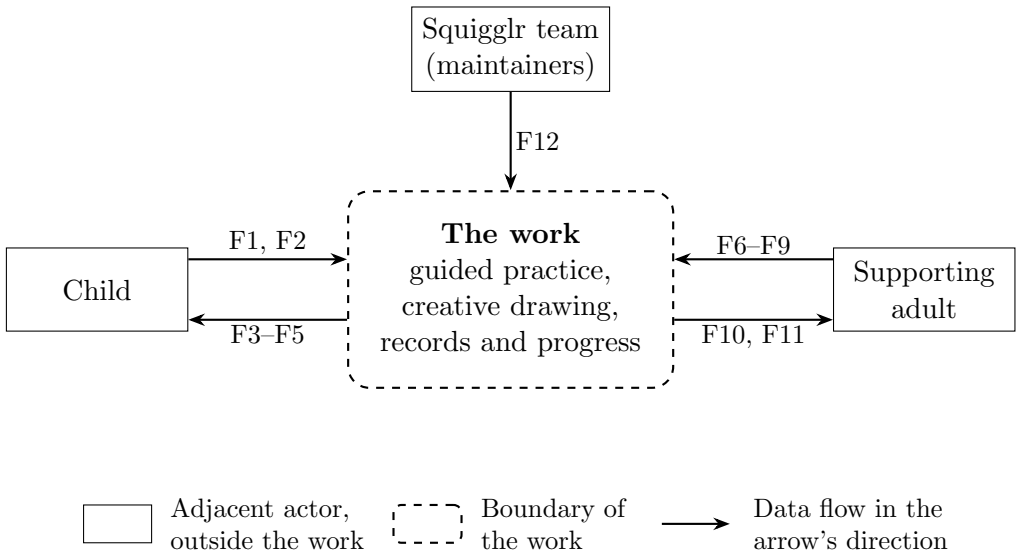


Figure 1: Context of the work. Each arrow carries the data flows named on it, which Table 11 defines.

Table 11: Data flows across the boundary of the work

Flow	From	To	Data
F1	Child	The work	Strokes drawn for an attempt or in the creative space.
F2	Child	The work	In the creative space, a request to turn refinement off or to restore the original drawing.
F3	The work	Child	Demonstrations and tracing guides for each stage.
F4	The work	Child	Feedback on an attempt, shown on screen and, unless muted, as audio.
F5	The work	Child	The refined drawing in the creative space.
F6	Supporting adult	The work	A child's profile details (first name and a username) and adult-set values (BR-001, BR-008).
F7	Supporting adult	The work	The lesson and starting stage chosen for a child, and the start of that lesson (BR-003).
F8	Supporting adult	The work	A decision on a suggestion to go back a stage: accept or decline.

Flow	From	To	Data
F9	Supporting adult	The work	Changes to a child's settings (accessibility, audio, whether refinement is on, adult-set values), or a request to restore an original drawing.
F10	The work	Supporting adult	A suggestion to go back a stage after repeated difficulty (BR-003).
F11	The work	Supporting adult	Progress summaries and the attempt records behind them.
F12	Squigglr team	The work	The lesson catalogue, the ranges of the adult-set values, and the fixed values (BR-009).

6.3 Work Partitioning

Table 12 partitions the work by business event: a happening outside the work, or a condition reached within it, that the work must respond to. Each event's inputs and outputs are data flows from Table 11, and each event has one business use case (BUC) in Section 6.4, with the same number.

Table 12: Business events

ID	Business event	Input	Output	BUC
BE-001	An adult wants a child to start practising with Squigglr.	F6	–	BUC-001
BE-002	An adult decides what the child should practise next.	F7	–	BUC-002
BE-003	An adult starts a guided lesson for the child.	F1, F7	F3, F4	BUC-003
BE-004	A child fails `DIFFICULTY_STREAK` attempts in a row at one stage without improving.	F8	F10	BUC-004
BE-005	A child wants to draw freely.	F1, F2, F9	F5	BUC-005
BE-006	An adult wants to know how the child's practice is going.	–	F11	BUC-006
BE-007	A child's needs change, so an adult changes the child's settings.	F9	–	BUC-007

ID	Business event	Input	Output	BUC
BE-008	The team changes the lesson catalogue or the ranges of the adult-set values.	F12	–	BUC-008

6.4 Specifying a Business Use Case (BUC)

Each business use case below is the work's response to the business event with the same number. It describes the whole response, including the parts done by people, and names the product use cases (Table 13) that are Squigglr's part of it.

BUC-001 Add a child (BE-001)
The adult creates a profile for the child and chooses each adult-set value, using the example drawings for `STAGE_PASS`, or leaves the defaults (BR-008). The work records the profile and the values. Product use case: UC-001.

BUC-002 Choose the next practice (BE-002)
The adult judges what the child is ready for, from knowing the child and, once there is practice, from the records (ASM-006). The adult then chooses a lesson and its starting stage. Product use cases: UC-002, UC-004.

BUC-003 Practise a guided lesson (BE-003)
The adult starts the lesson. The child watches the demonstration and draws an attempt at each stage. The work checks each attempt against the stage's criteria, gives feedback, and moves the child on or offers a retry. The lesson ends when the child finishes it or reaches `ATTEMPT_LIMIT`, or `ATTEMPT_LIMIT_MAX` when extended, and every attempt is recorded (BR-002). Product use cases: UC-003, UC-008.

BUC-004 Respond to repeated difficulty (BE-004)
The work suggests going back a stage. The adult decides whether to accept; a declined or ignored suggestion changes nothing (BR-003). Product use case: UC-005.

BUC-005 Draw freely (BE-005)
The child draws and edits in the creative space. The work refines the

drawing a little and keeps it editable. The child or the adult can turn refinement off or restore the original (BR-005; SRC-009.01). Product use cases: UC-006, UC-007.

BUC-006 Review progress (BE-006)
The adult reads a summary of the child's practice, opens the attempts behind it, and uses what they see to decide on further practice (BUC-002). The work gives no diagnosis or judgement of the child (BR-006). Product use case: UC-009.

BUC-007 Change a child's settings (BE-007)
The adult opens the child's history or settings and changes accessibility, audio or adult-set values. The work accepts only values within their ranges (BR-008). Product use case: UC-010.

BUC-008 Maintain lessons and ranges (BE-008)
The team adds or changes lessons, or changes the ranges of the adult-set values, using adult-made drawings (BR-009). Squigglr has no product use case for this event; the maintenance requirements cover it.

7 Business Data Model and Data Dictionary

7.1 Business Data Model

Insert your content here.

7.2 Data Dictionary

Insert your content here.

8 The Scope of the Product

8.1 Product Boundary

Insert your content here.

8.2 Product Use Case Table

Table 13 lists the ten product use cases, UC-001 to UC-010. Each row names the actor who starts or carries out the use case, the trigger that starts it, what the actor wants when it ends (the user goal) and the project goals it serves (Table 2). Squigglr is named as the actor where the product starts the use case itself. The identifiers are fixed and never renumbered; requirements trace to them, and Section 8.3 gives each use case's full flows. Names such as `STAGE_PASS` are named constants defined in the constants register (Table 10).

Table 13: Product use cases

ID	Actor: use case	Trigger	User goal	Goals
UC-001	*Supporting adult:* create a child's profile	The adult adds a child to Squigglr.	The child's profile exists, and every adult-set value is within its range, or at the middle of the range if the adult did not choose one.	G.3
UC-002	*Supporting adult:* choose a lesson	The adult chooses a lesson for a child.	The chosen lesson is ready for the adult to start.	G.1, G.3
UC-003	*Child:* practise a guided lesson	An adult starts a lesson for the child.	The child moves to the next stage after an attempt meets `STAGE_PASS`, or retries; the lesson ends when the child finishes it or reaches `ATTEMPT_LIMIT`, or `ATTEMPT_LIMIT_MAX` when extended (BR-002), and the furthest stage reached is saved.	G.1, G.2, G.3
UC-004	*Supporting adult:* start the child at a later stage	The adult judges that the child needs less help in a lesson.	The lesson starts at the stage the adult chose.	G.3
UC-005	*Supporting adult:* respond to a suggestion to go back a stage	Squigglr suggests an earlier stage after `DIFFICULTY_STREAK` failed attempts in a row.	The stage changes only if the adult accepts; a declined or ignored suggestion changes nothing.	G.3

ID	Actor: use case	Trigger	User goal	Goals
UC-006	*Child:* create and edit artwork	The child opens the creative space.	The child's drawing is saved and stays editable.	G.4
UC-007	*Squigglr:* refine artwork and restore the original	The child finishes a drawing in the creative space.	The drawing is refined within `REFINE_TOLERANCE` and stays editable; refinement can be turned off; the original strokes can be restored; a failed refinement leaves the drawing unchanged.	G.4
UC-008	*Squigglr:* record an attempt	The child ends an attempt at a lesson stage.	The record keeps the stage, every predefined difficulty observed, later corrections and the outcome.	G.5
UC-009	*Supporting adult:* review progress	The adult opens a child's progress summary.	The adult sees the child's progress over time, with attempts made without a guide separated from attempts made with one, and can open the attempts behind it.	G.6
UC-010	*Supporting adult:* review history and change settings	The adult opens a child's history or settings.	Only an authorised adult sees the history; the accessibility and audio settings, whether refinement is on, and the adult-set values are changed, and each adult-set value stays within its range.	G.3, G.6

8.3 Individual Product Use Cases (PUC's)

Insert your content here.

9 Functional Requirements

Functional requirements are grouped by externally observable product behaviour. Each requirement uses the identifier, fit-criterion and traceability conventions defined in the Introduction.

9.1 Learning Activity Requirements

This subsection specifies the externally observable behaviour of guided lessons. It builds on G.1–G.3, CON-003, CON-007 and BR-002–BR-009, with source statements in Appendix 29. The core catalogue covers lines, shapes and simple drawings; letter tracing is deferred (ASM-010, LC-001). Delivery-phase identifiers follow the conventions in the Introduction and the phase plan in Section 21.

Within-lesson progression and adult control. A supporting adult selects each lesson and its starting stage and authorizes skips or backward changes (G.3, CON-007). Within the selected lesson, a passing drawing attempt automatically advances to the next guidance stage (G.2, BR-002). Passing the independent stage completes the current lesson; Squigglr does not automatically select another lesson.

Lesson content and guidance. FR-LRN-001 — Approved lesson catalogue

Requirement: Squigglr shall offer guided lessons for the approved catalogue of lines, basic shapes and simple drawings, excluding letter-tracing activities from the core catalogue.
Rationale: These are the agreed categories for structured practice; letters are deferred (G.1, ASM-010). The lesson catalogue is maintained separately from the stage rules (LC-001).
Fit criterion: Inspect the adult-visible lesson catalogue and verify that every approved catalogue entry is selectable, classified as a line, shape or simple drawing, and has the stages in FR-LRN-002; verify that no letter activity is required to complete the core catalogue.
Priority: Must
Phase: PH-1
Sources: G.1; SRC-001; SRC-009.04

Traceability: ASM-010; LC-001; UC-002

FR-LRN-002 — Ordered guidance stages

Requirement: Squigglr shall make the following ordered guidance stages available for each guided lesson: demonstration, full-outline tracing, endpoint-only guidance and independent drawing.
Rationale: The activity described to specialists removes guidance in this order (G.1–G.2).
Fit criterion: For every approved lesson, inspect all four stages in the specified order; confirm that the stage chosen by an adult is the stage presented when practice begins.
Priority: Must
Phase: PH-1
Sources: G.1–G.2; SRC-001; SRC-005
Traceability: BR-003; UC-003; UC-004

FR-LRN-003 — Demonstration

Requirement: Squigglr shall show an example of the intended drawing during a lesson's demonstration stage without treating that demonstration as a learner attempt.
Rationale: A demonstration provides an example before guided practice and must not inflate the learner's records.
Fit criterion: Start each approved lesson in its demonstration stage; verify that the intended drawing can be viewed and that watching it adds no learner attempt or passed-stage record.
Priority: Must
Phase: PH-1
Sources: G.1; SRC-001; SRC-005
Traceability: BR-004; UC-003; UC-008

FR-LRN-004 — Full-outline guidance

Requirement: Squigglr shall display the lesson's complete reference outline during the full-outline tracing stage and accept stylus strokes drawn against it.
Rationale: Outline tracing is the first guided drawing stage in the approved progression.
Fit criterion: For each lesson, enter the full-outline stage, verify that the complete reference shape is visible, and draw an attempt using stylus input.

Priority: Must
Phase: PH-1
Sources: G.1; SRC-001; SRC-005
Traceability: CON-003; UC-003

FR-LRN-005 — Endpoint-only guidance

Requirement: Squigglr shall show the specified start and end points, without the full reference outline, during the endpoint-only stage and accept stylus strokes between those points.
Rationale: The middle drawing stage reduces visual assistance while retaining the destination cues.
Fit criterion: For each lesson, enter endpoint-only guidance; verify that the relevant endpoints are shown, the full outline is absent, and stylus drawing is accepted.
Priority: Must
Phase: PH-1
Sources: G.1–G.2; SRC-001; SRC-005
Traceability: CON-003; UC-003

FR-LRN-006 — Independent drawing

Requirement: Squigglr shall permit the drawing task to be attempted in the independent stage without displaying tracing outlines or endpoint guides.
Rationale: The final stage needs to be distinguishable from assisted practice (G.2, G.6).
Fit criterion: Enter the independent stage for each lesson; verify that neither an outline nor endpoint guides appear and that an attempt can still be made using the stylus.
Priority: Must
Phase: PH-1
Sources: G.2; SRC-001
Traceability: BR-004; UC-003; UC-008

Input, classification and feedback. FR-LRN-007 — Stylus stroke acquisition

Requirement: Squigglr shall accept stylus-drawn strokes as ordered sequences of input points for the active lesson attempt and preserve their order and stroke boundaries for subsequent classification.

Rationale: Stage checks and faithful attempt records require the original stroke structure (ASM-004, BR-004).
Fit criterion: Supply known one-stroke and multi-stroke stylus samples; verify that the captured point order, number of strokes and boundaries match the samples.
Priority: Must
Phase: PH-1
Sources: G.2, G.5; SRC-001; SRC-004
Traceability: CON-003; ASM-004; BR-004; UC-003; UC-008

FR-LRN-008 — Multi-stroke attempts

Requirement: Squigglr shall evaluate the collected strokes of a single attempt together rather than classify each stylus lift as a completed attempt.
Rationale: Lines and shapes can be composed of multiple strokes, and lifting a stylus alone does not establish that the task is finished.
Fit criterion: For a prepared two-stroke figure, lift the stylus after the first stroke and draw the second; confirm that one attempt is evaluated against the complete figure, not two independently completed attempts.
Priority: Must
Phase: PH-2
Sources: G.1–G.2; ASM-004; SRC-001
Traceability: FR-LRN-007; UC-003; UC-008

FR-LRN-009 — Interrupted attempt handling

Requirement: Squigglr shall distinguish an interrupted or unfinished drawing attempt from a completed drawing attempt, shall not record it as passed, and shall not consume an evaluable attempt from the stage limit.
Rationale: An input interruption or unfinished drawing is not evidence of a correctly completed task.
Fit criterion: Interrupt known partial attempts between and during strokes; verify that none is reported as passed, that the active stage does not advance, and that unfinished attempts are distinguishable from finalized attempts and that stage attempt counters are unchanged.
Priority: Must
Phase: PH-2
Sources: G.2, G.5; ASM-004; SRC-001
Traceability: FR-LRN-008; BR-004; UC-003; UC-008

FR-LRN-010 — Invalid and ambiguous input

Requirement: Squigglr shall withhold passing status from an attempt whose required stylus input is missing, invalid or too ambiguous to evaluate against the stage criteria, and shall not count such input toward the stage's evaluable-attempt or unsuccessful-attempt limits.
Rationale: Unreliable acquisition must not silently turn a malformed drawing into a successful result (ASM-004).
Fit criterion: Submit empty input, input with missing coordinate data and data for which the intended stroke order cannot be established; confirm that none passes, that a non-success status is available, and that the stage's attempt counters are unchanged.
Priority: Must
Phase: PH-1
Sources: G.2; ASM-004; BR-004; SRC-001
Traceability: FR-LRN-007; UC-003; H-AHM-001; H-AHM-003

FR-LRN-011 — Completeness classification

Requirement: Squigglr shall classify every evaluable drawing attempt as complete or incomplete relative to the selected lesson and guidance stage.
Rationale: Stage success must distinguish an unfinished line or missing side from a completed drawing (G.2).
Fit criterion: Evaluate adult-made or prepared examples with every required component present and with a deliberately missing side, segment or required endpoint; verify complete and incomplete classifications match the known examples.
Priority: Must
Phase: PH-1
Sources: G.2; BR-002; SRC-001
Traceability: FR-LRN-013; UC-003; UC-008

FR-LRN-012 — Guidance classification

Requirement: Squigglr shall classify each evaluated attempt as assisted when performed with outline or endpoint guidance and independent when performed without those guides.
Rationale: Adults need to distinguish guided practice from practice without a guide (G.6).

Fit criterion: Perform an evaluable attempt in each drawing stage; verify that outline and endpoint attempts are marked assisted and no-guide attempts are marked independent.
Priority: Must
Phase: PH-2
Sources: G.2, G.6; SRC-001
Traceability: FR-LRN-004–006; UC-008; UC-009

FR-LRN-013 — Pass classification

Requirement: Squigglr shall report an evaluable attempt as passing its stage only when the drawing is classified as complete and its result meets the current learner's `STAGE_PASS` criterion.
Rationale: A pass evaluates the submitted drawing against defined criteria, not the child's ability (G.2, BR-006).
Fit criterion: For prepared complete and incomplete drawings whose results lie below, at and above the adult-selected threshold, verify that only complete drawings meeting the threshold pass.
Priority: Must
Phase: PH-1
Sources: G.2; BR-002; BR-006; SRC-008
Traceability: FR-LRN-010–011; UC-003; H-AHM-004

FR-LRN-014 — Predefined difficulty feedback

Requirement: Squigglr shall present only pre-approved, stage-relevant feedback observations for evaluated attempts, including applicable missed tracing portions, incomplete lines or shapes and incorrect endpoints.
Rationale: Feedback must describe observable drawing difficulties and not generate personalized AI advice (BR-007).
Fit criterion: For each defined feedback category, evaluate a prepared matching drawing and verify that the corresponding approved observation appears; verify that no free-generated advice or diagnosis is shown.
Priority: Must
Phase: PH-2
Sources: G.1–G.2; BR-007; SRC-001; SRC-009.02
Traceability: FR-LRN-011; UC-003; H-AHM-005

FR-LRN-015 — Retry after non-passing attempts

Requirement: Squigglr shall allow the learner to retry a drawing stage fol-

lowing an attempt that does not pass, provided the stage has not reached its applicable attempt limit.
Rationale: G.1 requires opportunities to practise again without treating failure as a permanent stage result.
Fit criterion: Present a non-passing, evaluable sample before the limit and verify that another attempt can be started at the same drawing stage; repeat at the limit and verify that the lesson-end rule is followed instead.
Priority: Must
Phase: PH-1
Sources: G.1–G.3; BR-002; SRC-001
Traceability: FR-LRN-021; UC-003

Adult control, suggestions and stopping rules. FR-LRN-016 — Adult lesson and starting-stage selection

Requirement: Squigglr shall use the lesson and starting guidance stage selected by the supporting adult for the learner's session, including an adult-selected later starting stage.
Rationale: The adult controls initial level selection rather than the product assigning readiness (G.3, CON-007).
Fit criterion: Select each available starting stage as an adult, launch the lesson, and verify that the selected stage is used; check that a child-only practice action does not change the configured lesson or starting stage.
Priority: Must
Phase: PH-1
Sources: G.3; CON-007; SRC-005.01; SRC-006
Traceability: BR-003; UC-002; UC-004

FR-LRN-017 — Adult-selected practice parameters

Requirement: Squigglr shall apply the learner's adult-selected values for `STAGE_PASS`, `DIFFICULTY_STREAK`, `ATTEMPT_LIMIT`, `IMPROVE_WINDOW`, `ATTEMPT_EXTENSION` and `ATTEMPT_LIMIT_MAX`, using the documented midpoint default for any unset value and rejecting out-of-range or inconsistent selections.
Rationale: BR-008 requires learner-specific settings and rejects contradictory limit relationships.
Fit criterion: Test every setting at the minimum, default and maximum values in Table 10, plus one value outside each bound; verify that selected

values are used, defaults are applied correctly, and invalid combinations are rejected.
Priority: Must
Phase: PH-2
Sources: G.2–G.3; BR-008; SRC-008
Traceability: OI-05; UC-001; UC-010; H-AHM-007

FR-LRN-018 — Adult authorization for discretionary changes

Requirement: Squigglr shall require explicit authorization from the supporting adult before changing to another lesson, skipping a guidance stage or moving backward to an earlier stage; routine pass-based advancement to the next stage of the current lesson requires no additional adult action.
Rationale: Adults retain control of lesson selection and nonroutine guidance changes (G.3, CON-007), without interrupting the agreed forward progression on passing attempts (G.2, BR-002).
Fit criterion: Try selecting a new lesson, skipping ahead and applying a return suggestion without adult authorization, then repeat with authorization; verify that only authorized discretionary changes take effect. Verify independently that a passing attempt advances within the selected lesson without an additional adult action (FR-LRN-023).
Priority: Must
Phase: PH-2
Sources: G.2–G.3; CON-007; BR-002–003; SRC-005.01
Traceability: OI-04 (TA confirmation); FR-LRN-023; UC-004; UC-005; H-AHM-006

FR-LRN-019 — Difficulty suggestion

Requirement: Squigglr shall offer, without applying, an optional suggestion to return to an earlier drawing stage when the learner reaches `DIFFICULTY_STREAK` consecutive non-passing evaluable attempts at an eligible stage and the improvement rule does not suppress suggestions. When an adult declines or ignores the suggestion, another suggestion shall require a new full failure streak.
Rationale: A supporting adult should decide whether earlier guidance is appropriate (G.3, BR-003).
Fit criterion: Replay known failing sequences immediately below and at the selected streak threshold at both eligible stages, including sequences with fewer than twice the configured `IMPROVE_WINDOW` results and sequences with

qualifying improvement; verify that insufficient history does not delay the first suggestion, that qualifying improvement suppresses it, and that a declined suggestion reappears only after another full failure streak.
Priority: Must
Phase: PH-2
Sources: G.3; BR-003; SRC-005.01; SRC-008
Traceability: FR-LRN-018; FR-LRN-020; UC-005

FR-LRN-020 — Improvement-based extension

Requirement: Squigglr shall suppress a difficulty suggestion while the learner qualifies as improving, and, when the unsuccessful-attempt count first reaches `ATTEMPT_LIMIT`, grant up to `ATTEMPT_EXTENSION` additional evaluable stage attempts if the average result of the most recent `IMPROVE_WINDOW` attempts exceeds the average of the preceding equally sized window by at least `IMPROVE_MARGIN`, subject to `ATTEMPT_LIMIT_MAX`. The extension shall be granted at most once per stage and expire after the allotted additional evaluable attempts, even if results continue to improve.
Rationale: The agreed improvement rule reduces premature interruption while retaining a hard cap (G.3, BR-002).
Fit criterion: Evaluate prepared improving and non-improving sequences exactly at the margin boundary and at the applicable limit; verify extension and suggestion suppression only for qualifying improvement, exactly the permitted number of extension attempts without regranting, and no attempt beyond the maximum.
Priority: Must
Phase: PH-2
Sources: G.3; BR-002–003; SRC-001; SRC-008
Traceability: FR-LRN-019; FR-LRN-021; OI-05; UC-003

FR-LRN-021 — Stage attempt limit and lesson end

Requirement: Squigglr shall end the active lesson when its unsuccessful-attempt count reaches `ATTEMPT_LIMIT` without a qualifying extension, or a granted extension is exhausted, or the allowed total-attempt count reaches `ATTEMPT_LIMIT_MAX`, and make the furthest stage reached available for the lesson record.
Rationale: The limits prevent indefinite repetition and preserve usable progress (G.3, G.5, BR-002).

Fit criterion: Replay sequences at the base limit and at the maximum, with and without qualifying improvement; verify that a granted extension expires after the allotted attempts, does not renew, and is capped by the maximum. Verify that the highest stage reached is included in the saved lesson outcome.
Priority: Must
Phase: PH-2
Sources: G.3, G.5; BR-002; SRC-001
Traceability: FR-LRN-015; FR-LRN-020; UC-003; UC-008

FR-LRN-022 — No learner evaluation or diagnosis

Requirement: Squigglr shall express lesson feedback and completion results as properties of the drawing attempt rather than as assessments, diagnoses or comparisons of children.
Rationale: The product is a practice tool, not an assessment instrument (BR-006).
Fit criterion: Inspect all guided-lesson feedback and completion outcomes, using prepared attempts; verify that each refers to the task, stage or observed strokes and that none assigns a diagnosis, ability label or inter-child ranking.
Priority: Must
Phase: PH-1
Sources: G.2; BR-006; SRC-001; SRC-008
Traceability: FR-LRN-013–014; UC-003; UC-009

FR-LRN-023 — Automatic within-lesson advancement

Requirement: Squigglr shall automatically advance from full-outline tracing to endpoint-only guidance, and from endpoint-only guidance to independent drawing, upon a passing attempt at the active stage; a passing independent drawing attempt completes the current lesson without starting a different lesson.
Rationale: G.2 and BR-002 require predictable reduction of guidance after success. Adult consent governs lesson selection, skips and backward moves, not normal pass-based progression within the selected lesson (CON-007).
Fit criterion: For a selected lesson, finalize passing attempts at its outline, endpoint and independent stages; verify the exact transitions T to E, E to I and I to completed, respectively, without an adult interaction. Verify that a non-passing attempt stays at its current stage, a different lesson is not started, and no stage is skipped automatically.
Priority: Must

Phase: PH-1
Sources: G.2–G.3; BR-002–003; CON-007; SRC-001
Traceability: FR-LRN-013; FR-LRN-016; FR-LRN-018; UC-003; UC-004; OI-04 (TA confirmation)

Source-to-requirement traceability. Table 14 provides source-side links for the most important learning-activity decisions; each requirement also links back to its evidence in its *Sources* field. The full system-wide traceability matrix is maintained by the owner of Section 28.

Table 14: Learning-activity evidence and derived requirements

Source or rule	Related learning requirements
G.1, SRC-001	FR-LRN-001, FR-LRN-002, FR-LRN-003, FR-LRN-004, FR-LRN-014, FR-LRN-015
G.2, BR-002	FR-LRN-005, FR-LRN-006, FR-LRN-011, FR-LRN-013, FR-LRN-020, FR-LRN-021, FR-LRN-023
G.3, BR-003, CON-007	FR-LRN-016, FR-LRN-018, FR-LRN-019, FR-LRN-023; OI-04
BR-008, SRC-008	FR-LRN-013, FR-LRN-017, FR-LRN-019, FR-LRN-020
ASM-004, BR-004	FR-LRN-007, FR-LRN-008, FR-LRN-009, FR-LRN-010
BR-006–007	FR-LRN-014, FR-LRN-022
SRC-005.01, SRC-005.05	FR-LRN-016, FR-LRN-018, FR-LRN-019
SRC-009.02, SRC-009.04	FR-LRN-001, FR-LRN-014

9.2 Formal Specification

The most safety-relevant learning behaviour is formalized here as a *labelled transition system*: a set of states and named events with rules describing when an event changes the state. The model specifies externally observable outcomes rather than a particular user interface or scoring algorithm. It covers guidance levels, attempt classification, optional suggestions, automatic

pass-based advancement within a lesson, and the unsuccessful-attempt stopping rule. Every variable used below has a type or domain; arrows indicate transitions *caused by the event above the arrow*, not automatic passage of time. This notation applies only to guided learning activities.

Domains and state. For each selected lesson ℓ and learner profile p, let

$$\begin{aligned} L &= \{D, T, E, I\}, & A &= \{T, E, I\}, \\ R &= \{\mathsf{complete}, \mathsf{incomplete}, \mathsf{invalid}\}, \\ M &= \{\mathsf{assisted}, \mathsf{independent}\}, \\ Q &= \{\mathsf{active}, \mathsf{suggested}, \mathsf{ended}\}. \end{aligned}$$

The levels D, T, E, I mean *demonstration*, full-outline *tracing*, *endpoint-only* and *independent* drawing, respectively. Only levels in A admit scored drawing attempts; D is a non-scored demonstration. A state is

$$\sigma = (\ell, p, l, q, n_f, n_a, k, H, s, z, x, f),$$

where $l \in L$ is the current guidance level of the adult-selected lesson, $q \in Q$ is the session status, $n_f \in \mathbb{N}_0$ counts non-passing evaluable attempts at this stage, $n_a \in \mathbb{N}_0$ counts finalized, evaluable attempts at this stage, $k \in \mathbb{N}_0$ is the consecutive non-passing streak, and $H = (r_1, \ldots, r_m)$ is the ordered list of stage-result values. The flag $s \in \{0, 1\}$ indicates a pending suggestion; $z \in \{0, 1\}$ indicates whether the single stage extension has been granted; $x \in \mathbb{N}_0$ counts evaluable attempts consumed *after* that grant, with $0 \leq x \leq e_p$ and $x = 0$ when $z = 0$. Finally, $f \in L$ is the furthest stage reached in the lesson. On a stage change (either normal pass-based progression or an authorized adult adjustment), the stage-specific counts, streak, history, pending suggestion and extension state reset to zero or empty; f retains the furthest stage reached so far. *No child ability estimate* is part of this state (BR-006).

Drawing input and attempt classification. An attempt is a finite ordered list of stylus strokes $X = (x_1, \ldots, x_j)$, where every x_i is an ordered list of reported points. Lifting the stylus closes one x_i, not the attempt. The terminal event Finalize(X) denotes the end of input collection for an attempt. This abstract event does not prescribe a particular user-interface control. A

stylus lift alone does not finalize the attempt; interruption before finalization cannot produce a passed result (FR-LRN-008–010).

Let $\mathrm{Valid}_{\ell,l}(X)$ mean that all input required to interpret the strokes is available, and let $\mathrm{Complete}_{\ell,l}(X)$ mean that every required part of the drawing is present according to the lesson's fixed stage criteria. These functions use the fixed completeness criteria of the selected lesson and stage. Stage-result values use the normalized 0–100 scale specified in Table 10; the actual scoring function is defined for each lesson's drawing criteria rather than as an assessment of a learner. Define

$$C_{\ell,l}(X) = \begin{cases} \mathsf{invalid} & \neg\,\mathrm{Valid}_{\ell,l}(X), \\ \mathsf{complete} & \mathrm{Valid}_{\ell,l}(X) \wedge \mathrm{Complete}_{\ell,l}(X), \\ \mathsf{incomplete} & \text{otherwise.} \end{cases}$$

The guidance classification uses the level of the actual attempt:

$$G(l) = \begin{cases} \mathsf{assisted} & l \in \{T, E\}, \\ \mathsf{independent} & l = I. \end{cases}$$

Let $v_{\ell,l}(X) \in [0, 100]$ be the normalized stage result of an evaluable drawing. Its threshold is $\theta_p = \texttt{STAGE_PASS}(p)$, taken from the adult's approved profile value (or midpoint default under BR-008). The pass predicate is

$$\mathrm{Pass}(X, l, p) \iff C_{\ell,l}(X) = \mathsf{complete} \wedge v_{\ell,l}(X) \geq \theta_p.$$

A missing, malformed or interrupted attempt never satisfies this predicate. Non-passing but evaluable attempts contribute to the unsuccessful-attempt count. Invalid or interrupted input is never interpreted as success and does not contribute to a stage's evaluated-attempt or unsuccessful-attempt counters. The presented feedback is selected from the fixed set of observations in BR-007 rather than generated advice.

Improvement and suggestion predicates. Use the constants defined in Section 4.2:

$$\begin{aligned} d_p &= \texttt{DIFFICULTY_STREAK}(p), & b_p &= \texttt{ATTEMPT_LIMIT}(p), \\ w_p &= \texttt{IMPROVE_WINDOW}(p), & e_p &= \texttt{ATTEMPT_EXTENSION}(p), \\ h_p &= \texttt{ATTEMPT_LIMIT_MAX}(p), & \delta &= \texttt{IMPROVE_MARGIN}. \end{aligned}$$

For $H = (r_1, \ldots, r_m)$, define the total improvement predicate

$$\text{Improving}(H, p) = \begin{cases} \mathsf{false}, & m < 2w_p, \\ \left(\dfrac{1}{w_p} \displaystyle\sum_{i=m-w_p+1}^{m} r_i - \dfrac{1}{w_p} \sum_{i=m-2w_p+1}^{m-w_p} r_i \geq \delta \right), & m \geq 2w_p. \end{cases}$$

Insufficient history therefore does not count as improvement and does not delay a difficulty suggestion. For the default $w_p = 2$, the first comparison uses four evaluable attempts; a shorter history still allows the suggestion at the selected failure-streak threshold.

For an eligible return to a preceding guidance level, the suggestion trigger is

$$\text{SuggestBack}(\sigma) \iff q = \mathsf{active} \wedge s = 0 \wedge l \in \{E, I\} \wedge k \geq d_p \wedge \neg\, \text{Improving}(H, p).$$

The rule applies to drawing levels E and I; T has no preceding drawing level. A suggestion changes q to $\mathsf{suggested}$ and s to 1, but *does not change* l. Declining or ignoring a suggestion resets the streak counter k to zero, preventing an immediate re-offer; a new suggestion requires another d_p consecutive failures.

State-transition rules. Table 15 gives the level-changing and stage-stopping rules. Here σ' is the state after the event, "unchanged" means the active lesson and level remain exactly as before, and AdultAccept is a verified adult action rather than any child drawing input. Likewise, AdultSelect is the authorized action selecting a lesson and its starting level. A granted extension sets $z = 1$ once, with $x = 0$; each subsequently finalized evaluable attempt increments x. The extension cannot be granted again at this stage even if later scores qualify as improvement. After finalization, a passing result takes priority over stop and suggestion rules. Otherwise, reaching the base failure limit first triggers either the single extension grant (if improving and still below the hard cap) or lesson end. After a grant, exhausting $x = e_p$ or reaching $n_a = h_p$ ends the lesson. A suggestion is offered only when none of these higher-priority rules ends the lesson. For pass-based forward transitions define $\text{Next}(T) = E$ and $\text{Next}(E) = I$; I has no successor within its lesson. Passing the last stage ends *that* lesson.

Table 15: Guided-lesson state transitions and authorization

Event	**Guard**	**Required state change or output**
Adult selects lesson / starting level (AdultSelect)	Supporting adult is authorized	Set (ℓ, l) to adult selection, $q = \mathsf{active}$; reset stage counters, history, suggestion and extension state; keep any previously persisted lesson record.
Begin practice after demonstration	$q = \mathsf{active}$, $l = D$; child continues after viewing the example	Set $l = T$ within the same lesson; initialize drawing-stage counters. This does not start another lesson or require new adult authorization.
Finalize evaluable attempt	$l \in A$, $q = \mathsf{active}$	Classify $C_{\ell,l}(X)$ and $G(l)$; report pass/fail using the stage threshold; increment n_a and, for non-pass, n_f, k. If $z = 1$, increment x for this additional evaluable attempt. On pass, record the result and apply the prescribed forward-stage or lesson-end transition below.
Interrupted or invalid input	Missing finalization or $C_{\ell,l}(X) = \mathsf{invalid}$	Do not pass or change level. Neither n_a nor n_f is incremented, and no successful stage transition is permitted.
Offer suggestion	SuggestBack(σ)	Set $s = 1$, $q = \mathsf{suggested}$; leave ℓ, l unchanged.
Ignore or decline suggestion	$s = 1$	Clear s, reset $k = 0$, return q to active; leave ℓ, l unchanged and retain the attempt history H and cumulative stage counts.

Event	**Guard**	**Required state change or output**
Adult accepts suggestion	$s = 1$ and an authorized adult chooses a preceding level	Activate the chosen earlier level, clear s, reset stage counts; retain the lesson's furthest stage reached.
End at limit	The finalized attempt did not pass and either $z = 0 \land n_f \geq b_p \land (\neg\,\text{Improving}(H, p) \lor n_a \geq h_p)$, or $z = 1 \land (x \geq e_p \lor n_a \geq h_p)$	Set $q = \mathsf{ended}$; report the furthest stage reached for saving; no more attempts in the ended session. A pass takes priority over this stop condition.
Extend at base limit	$z = 0$, $n_f \geq b_p$, $\text{Improving}(H, p)$, $n_a < h_p$, and the finalized attempt did not pass	Set $z = 1$, $x = 0$; allow at most e_p additional evaluable attempts, subject to $n_a < h_p$. Do not regrant this extension or offer a conflicting return suggestion.
Passing a nonfinal drawing stage	$\text{Pass}(X, l, p)$ and $l \in \{T, E\}$	Save the passed attempt at its original stage; automatically set $l' = \text{Next}(l)$ without adult action. Reset stage-specific counters and update f; leave ℓ unchanged.
Passing the final drawing stage	$\text{Pass}(X, I, p)$	Save the passed independent attempt, set $q' = \mathsf{ended}$ and $f' = I$; complete the current lesson without choosing another lesson.

Invariants and verification obligations. For every reachable session state and every adult- or learner-generated event, the model requires:

I1: $\ell' \neq \ell \Longrightarrow$ AdultSelect;

I2: suggestion ignored or declined $\Longrightarrow (\ell', l') = (\ell, l)$;

I3: $C_{\ell,l}(X) \neq \mathsf{complete} \Longrightarrow \neg\,\mathrm{Pass}(X, l, p)$;

I4: pass/feedback never substitutes refined strokes for actual guided strokes;

I5: $n_a \leq h_p \ \wedge \ 0 \leq x \leq e_p \wedge (z = 0 \Rightarrow x = 0)$;

I6: $\mathrm{Pass}(X, T, p) \Longrightarrow (\ell', l') = (\ell, E)$;

I7: $\mathrm{Pass}(X, E, p) \Longrightarrow (\ell', l') = (\ell, I)$;

I8: $\mathrm{Pass}(X, I, p) \Longrightarrow q' = \mathsf{ended} \wedge \ell' = \ell$.

Outside ordinary pass-based progression and the demonstration-to-practice transition, a change to the active stage requires an authorized adult selection or acceptance (CON-007). Invariants I1 and I6–I8 ensure that a new lesson is never selected automatically and that passing an attempt advances only within the current lesson. I2–I5 cover suggestions, invalid input, faithful records, bounded stage attempts and the one-time extension. These rules are verified using prepared adult-made attempt sequences (CON-005–006), including threshold cases with insufficient improvement history, declined suggestions, a full extension and the hard cap in Table 10.

9.3 Creative Drawing Requirements

Insert your content here.

9.4 Attempt Record Requirements

Insert your content here.

9.5 Progress Summary Requirements

Insert your content here.

10 Look and Feel Requirements

10.1 Appearance Requirements

Insert your content here.

10.2 Style Requirements

Insert your content here.

11 Usability and Humanity Requirements

11.1 Ease of Use Requirements

Insert your content here.

11.2 Personalization and Internationalization Requirements

Insert your content here.

11.3 Learning Requirements

Insert your content here.

11.4 Understandability and Politeness Requirements

Insert your content here.

11.5 Accessibility Requirements

Insert your content here.

Note for the section owner (from #87): add a requirement that Squigglr shall meet selected Web Content Accessibility Guidelines (WCAG) 2.2 success criteria; the exact set is TBD and should be criteria the team can check without children, such as colour contrast, text alternatives for icons and audio that can be paused or muted. Full WCAG conformance is a stretch goal. Source: SRC-005, P2, 39:26. Related advice: icon-based navigation

that children can use without reading (SRC-005, P1, 34:13) and audio feedback with a mute option (SRC-005, P2, 37:39). Add a WCAG 2.2 citation to refs/References.bib. The standard itself goes in Standards Compliance Requirements.

12 Performance Requirements

12.1 Speed and Latency Requirements

Insert your content here.

12.2 Safety-Critical Requirements

This subsection holds safety requirements that come from the Hazard Analysis and do not fit another section. Each lists the hazards it mitigates.

SAF-RAY-001 Squigglr shall allow lesson choice, starting-stage choice, learner switching and acceptance of a go-back suggestion only after an adult check that a child cannot complete by tapping icons, and shall keep a pending suggestion, without blocking practice, until an adult accepts or declines it.
Rationale: children should be able to practise without constant adult help (SRC-005, P1, 34:13), but progression belongs to adults (CON-007; SRC-005, P1, 28:37). The form of the check is `ADULT_CHECK`, TBD.
Fit criterion: in prepared walkthroughs, no adult-only action completes without passing the adult check, and a pending suggestion changes nothing until an adult acts.
Hazards: H-RAY-001, H-RAY-006.

SAF-RAY-002 Before a lesson starts, Squigglr shall check the device and browser against `SUPPORTED_DEVICES` (TBD) and, if they do not match, show an adult-readable notice that results may be unreliable.
Rationale: the team does not control the device (CON-001–CON-003).
Fit criterion: on each prepared unsupported combination the notice appears before the first attempt; on supported ones it does not.
Hazards: H-RAY-002.

SAF-RAY-003 Squigglr shall keep each completed attempt on the device until the server confirms it is saved, and shall tell the adult which attempts could not be saved.
Rationale: offline access is not promised (SRC-005, Moustafa, 6:03), and G.5 needs complete records.
Fit criterion: with the connection cut during or after prepared attempts, no confirmed attempt is lost and every unsaved attempt is reported to the adult.
Hazards: H-RAY-002.

SAF-RAY-004 No record, summary or export shall show a diagnostic label, a judgement of the child's abilities or a comparison with other children, and every summary and export shall state that it describes practice only and is not an assessment or diagnosis.
Rationale: Squigglr makes no diagnosis (SRC-001; SRC-005, Moustafa, 6:03), and adults reading records may have no training.
Fit criterion: inspection of every summary and export template finds none of the excluded content and finds the statement present.
Hazards: H-RAY-003.

SAF-RAY-005 Squigglr shall end a lesson at `ATTEMPT_LIMIT`, or at `ATTEMPT_LIMIT_MAX` when extended, without any action from the child, and shall keep the progress saved up to that point.
Rationale: the child is not expected to exit a lesson, and no adult may be watching (CA-001), so repeated failure must end on its own (G.3).
Fit criterion: in prepared sequences of failed attempts, each lesson ends at the limit in use with no child input, and the saved record matches the attempts made.
Hazards: H-RAY-004.

SAF-RAY-006 Squigglr shall record each lesson that ends at `ATTEMPT_LIMIT` or `ATTEMPT_LIMIT_MAX`, and each improvement extension, show the adult how many times these happened for each lesson, and keep earlier attempts when a lesson is restarted.
Rationale: a restart could otherwise hide repeated failure from the adult.
Fit criterion: in prepared sequences that reach the limit and then restart, the count and all earlier attempts appear in the adult view.
Hazards: H-RAY-005.

SAF-RAY-007 Squigglr shall identify the active learner on every practice screen in a form a child who cannot read can recognize, and shall save every attempt and suggestion to that learner only.
Rationale: school devices may be shared by several children (SRC-004).
Fit criterion: in prepared multi-learner sessions on one device, every attempt and suggestion is saved under the learner shown when it was made.
Hazards: H-RAY-006.

SAF-RAY-008 Squigglr shall save a child's adult-set values (Table 10) only within their team-set ranges, with `ATTEMPT_LIMIT` larger than `DIFFICULTY_STREAK` and `ATTEMPT_LIMIT_MAX` at least `ATTEMPT_LIMIT`; shall use the middle of each range when the adult does not choose; shall allow changes only after the adult check; and shall show the values in use alongside the child's records.
Rationale: the supervisor advised that the adult chooses these values (SRC-008); a wrong value can stop a child passing a stage they can draw, and a child must not change it.
Fit criterion: in prepared walkthroughs, every out-of-range value and every `ATTEMPT_LIMIT` not larger than `DIFFICULTY_STREAK` and every `ATTEMPT_LIMIT_MAX` below `ATTEMPT_LIMIT` is refused, an unset value equals the midpoint, and no change completes without the adult check.
Hazards: H-RAY-007.

SAF-AHM-001 Squigglr shall prevent duplicate reports of the same stylus input event from being counted more than once during evaluation or recording of an attempt, while preserving legitimate separately drawn strokes, including overlapping strokes.
Rationale: duplicate input events can inflate drawing coverage, produce incorrect pass or failure decisions, and cause saved records to differ from what the learner actually drew (H-AHM-002; BR-004).
Fit criterion: replay a known one-stroke or multi-stroke input sequence with and without duplicated input events; verify that the recorded points, classification and pass result are identical. Replay two separate deliberately overlapping strokes and verify that both are preserved.
Related SRS requirements: FR-LRN-007, FR-LRN-008.
Hazards: H-AHM-002.

SAF-AHM-002 Squigglr shall preserve the relative positions and order of

stylus points within the active drawing surface when the canvas size or supported device orientation changes, and shall reject input whose coordinates cannot be mapped reliably to that surface.
Rationale: an incorrect coordinate mapping can distort a learner's drawing, produce an inaccurate passing decision, and corrupt stored attempts (H-AHM-003; BR-004).
Fit criterion: replay the same prepared stroke sequences before and after resizing the drawing surface or changing supported device orientation; verify equivalent relative point positions, stroke order and pass classifications. Supply invalid or unmappable coordinate values and verify that none is classified as passing or counted as an evaluable attempt.
Related SRS requirements: FR-LRN-007, FR-LRN-010.
Hazards: H-AHM-003.

12.3 Precision or Accuracy Requirements

Insert your content here.

12.4 Robustness or Fault-Tolerance Requirements

Insert your content here.

12.5 Capacity Requirements

Insert your content here.

12.6 Scalability or Extensibility Requirements

Insert your content here.

12.7 Longevity Requirements

Insert your content here.

13 Operational and Environmental Requirements

13.1 Expected Physical Environment

Insert your content here.

13.2 Wider Environment Requirements

Insert your content here.

13.3 Requirements for Interfacing with Adjacent Systems

Insert your content here.

13.4 Productization Requirements

Insert your content here.

13.5 Release Requirements

Insert your content here.

14 Maintainability and Support Requirements

14.1 Maintenance Requirements

Insert your content here.

14.2 Supportability Requirements

Insert your content here.

14.3 Adaptability Requirements

Insert your content here.

15 Security Requirements

15.1 Access Requirements

Insert your content here.

15.2 Integrity Requirements

Insert your content here.

15.3 Privacy Requirements

Insert your content here.

15.4 Audit Requirements

Insert your content here.

15.5 Immunity Requirements

Insert your content here.

16 Cultural Requirements

Insert your content here.

17 Compliance Requirements

17.1 Legal Requirements

Insert your content here.

17.2 Standards Compliance Requirements

Insert your content here.

Note for the section owner (from #87): list WCAG 2.2 as the accessibility standard Squigglr follows for selected success criteria (see Accessibility Requirements). It is not legally required for this project; the domain specialists

recommended it as a starting point (SRC-005, P2, 39:26). Full conformance is a stretch goal.

18 Open Issues

This section lists questions that are not yet settled and could change the requirements. Each issue has an identifier (OI-##), an owner, the decision or evidence needed to close it, the goals, constraints or requirements it affects, and a target date. Requirement identifiers are linked here as the requirements are written. Target dates come from the SRS schedule (review copy Oct. 17, 2026; submission Oct. 19, 2026) or from the course deadlines in Table 8 (CON-004) where an issue cannot be closed before submission.

Table 16: Open issues

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-01	*School and shared devices.* School devices may be shared by several children (SRC-004) and are often managed centrally (CON-002). One adult account holds several learner profiles, and an adult switches between them (BR-001). Still needed: whether an adult signs in on each shared device, and how a device is left when a session ends.	G.5, G.6, CON-001, CON-002; requirement IDs to be linked when written	Shamil Canbolat	Oct. 17, 2026
OI-02	*Privacy of practice records.* The supervisor's advice to collect no information about children (SRC-004, #21) concerns personal information; storing practice attempts is acceptable. Still needed: the privacy requirements for attempt records, including who can see them, whether and how they can be shared, and how they are deleted.	G.5, G.6, CON-005; privacy requirement IDs to be linked when written	Shamil Canbolat	Oct. 17, 2026

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-03	*Scope of G.4.* A peer reviewer suggested moving creative drawing and refinement to the stretch goals (#70). The team may adopt the change or reply on the issue explaining why G.4 stays a core goal (SRC-006). Still needed: the reply on the issue and, if G.4 stays, a supported set of items small enough for the schedule.	G.4	Muhammad Huzaifah	Oct. 17, 2026
OI-04	*TA confirmation of revised G.3.* The team distinguishes adult selection of lessons, starting stages, skips and backward moves from automatic advancement after a passing attempt within the selected lesson (G.2, BR-002, CON-007). G.3 permits an optional suggestion to go back a stage after `DIFFICULTY_STREAK` failed attempts and a lesson end after `ATTEMPT_LIMIT`, held back while the child is improving (`IMPROVE_WINDOW`, `ATTEMPT_EXTENSION`). The TA agreed to the adult-control revision in principle and asked that the updated PSG be merged after marking the submitted version (SRC-006). Still needed: confirmation of the rule as clarified, not a new choice between automatic and adult-authorized forward progression.	G.3, CON-007	Shamil Canbolat	Oct. 17, 2026

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-05	*Verification of learning thresholds.* Section 4.2 defines the ranges and defaults for the six adult-configurable values and the fixed `IMPROVE_MARGIN`. Verify the selected ranges with adult-made drawings, provide examples illustrating the endpoints of the `STAGE_PASS` range, and confirm that the drawing-evaluation method produces results on the specified 0–100 scale (SRC-004, SRC-008). No child testing is permitted.	G.2, G.3; FR-LRN-017, FR-LRN-020, FR-LRN-021	Ahmed Elzaria	Nov. 2, 2026 (V&V Plan)
OI-06	*Data isolation between institutions.* A peer reviewer asked how records from different schools or centres are kept apart (#75). Still needed: a decision on whether Squigglr supports separate institutions in this project and, if so, the access and privacy requirements that keep their records apart.	G.5, G.6; access and privacy requirement IDs to be linked when written	Shamil Canbolat	Oct. 17, 2026
OI-07	*Budget.* No budget has been set (Budget Constraints), and external services may have costs. Still needed: the TA's confirmation of whether any budget is available.	G.4 (refinement service)	Shamil Canbolat	Oct. 17, 2026
OI-08	*Selected WCAG 2.2 success criteria.* The specialists recommended starting with WCAG (SRC-005, P2, 39:26); the set of criteria Squigglr follows is TBD. The supervisor advised taking specific WCAG ideas as non-functional requirements rather than as a constraint (SRC-008). Still needed: a list of criteria the team can check without children.	G.1; accessibility and standards requirement IDs to be linked when written	Moustafa Moustafa (Usability); Shamil Canbolat (Compliance)	Oct. 17, 2026

ID	Issue and decision or evidence needed	Affects	Owner	Target
OI-09	*Supported devices and browsers.* CON-003 leaves the device and browser combinations TBD. Still needed: the list of combinations the team will support and test with a stylus.	CON-001, CON-003	Shamil Canbolat	Nov. 2, 2026 (V&V Plan)

19 Off-the-Shelf Solutions

19.1 Ready-Made Products

Insert your content here.

19.2 Reusable Components

Insert your content here.

19.3 Products That Can Be Copied

Insert your content here.

20 New Problems

20.1 Effects on the Current Environment

Insert your content here.

20.2 Effects on the Installed Systems

Insert your content here.

20.3 Potential User Problems

Insert your content here.

20.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

Insert your content here.

20.5 Follow-Up Problems

Insert your content here.

21 Tasks

21.1 Project Planning

Insert your content here.

21.2 Planning of the Development Phases

Insert your content here.

22 Migration to the New Product

22.1 Requirements for Migration to the New Product

Insert your content here.

22.2 Data That Has to be Modified or Translated for the New System

Insert your content here.

23 Costs

Insert your content here.

24 User Documentation and Training

24.1 User Documentation Requirements

Insert your content here.

24.2 Training Requirements

Insert your content here.

25 Waiting Room

Insert your content here.

26 Ideas for Solution

Insert your content here.

27 Likely and Unlikely Changes

This section is added to the Volere template because the course asks for it. It records which parts of the requirements are expected to change, so that the design can keep them easy to change, and which are not expected to change, so that the design can rely on them. Each change has an identifier: LC-### for a likely change and ULC-### for an unlikely one.

27.1 Likely Changes

Table 17: Likely changes

ID	Change and why it is likely	Affects	How the design can hide it
LC-001	The lesson catalogue (which lines, shapes and drawings, in what order) may change, for example to add letters after shapes (SRC-009.04).	G.1, G.2	Lessons and their stages can be stored as data that the lesson logic reads, so adding or reordering a lesson changes no code.
LC-002	The adult-configurable ranges in Table 10 may be recalibrated after adult-made verification (OI-05), and adults may change their chosen values at any time.	G.2, G.3, CON-007	Each range is defined in one place, each child's value is stored in their profile, and the activity logic reads only the profile value.
LC-003	The supported device and browser combinations are TBD and will change as browsers and tablets change (OI-09).	CON-001, CON-003	Drawing input can be read through one input module that turns browser pointer events into strokes; the rest of the product sees only strokes.
LC-004	The service that refines creative drawings may change because of cost (OI-07), quality or availability. The refinement is planned to use a lightweight AI model (SRC-005, Moustafa, 6:03).	G.4	Refinement can sit behind one interface that takes stroke coordinates and returns adjusted coordinates at the same scale, so a different provider replaces one module.

ID	Change and why it is likely	Affects	How the design can hide it
LC-005	Feedback wording, colours and audio will change: the specialists advised designing them with educators and occupational therapists, bit by bit (SRC-005, P1, 36:26; P2, 37:39).	G.1	Feedback text, colours and sounds can be kept as content separate from the rules that decide which feedback to give.
LC-006	Finger input may be added later; it is a stretch goal, and the specialists suggested finger, large stylus and small stylus as possible difficulty levels (SRC-005, P1, 26:46).	CON-003, G.1	The input module of LC-003 can accept a new input type without changing the lesson logic or the records.
LC-007	Which recorded values each adult wants to see will change from adult to adult and over time.	G.6	Squigglr records every value and lets the adult choose what their view shows, so a new view needs no change to what is recorded.
LC-008	The way improvement is measured (average of recent attempts compared with earlier ones) may be replaced once it is tried.	G.3, BR-002	The measure is one function that reads the stored attempt results; the lesson logic only asks whether the child is improving.

27.2 Unlikely Changes

Table 18: Unlikely changes

ID	Fact not expected to change	Affects	Justification and source
ULC-001	Squigglr is delivered as a website, with no installation.	CON-001, CON-002	Imposed by the supervisor and TA because installing apps on school-managed devices is difficult (SRC-003, SRC-004). This may change if browser input proves unreliable (ASM-004).
ULC-002	Only supporting adults view and manage practice records; children do not.	G.5, G.6	The records exist to help adults understand practice (SRC-001), adults handle setup (SRC-005, P1, 34:13), and the records concern children, whose privacy the product must protect (OI-02).
ULC-003	An adult chooses the lesson, starting stage, and any skip or backward move. Passing an attempt advances to the next stage within the chosen lesson, never to a new lesson.	G.2, G.3, CON-007	Deciding which lesson or exceptional level a child needs automatically would require assessment outside the project scope (SRC-005, P1, 28:37; P2, 29:18); the TA agreed to the change (SRC-006).
ULC-004	Squigglr gives no diagnosis, no judgement of a child's abilities, no comparison with other children and no effectiveness claims.	G.2, G.6	Excluded from the start (SRC-001) and presented to the specialists as deliberately left out (SRC-005, Moustafa, 6:03); the team cannot judge a child's abilities (SRC-008), and effectiveness claims would need studies the project cannot run (CON-006).

ID	Fact not expected to change	Affects	Justification and source
ULC-005	In guided lessons, the record keeps the child's actual strokes; refinement happens only in the creative space, and the original can be restored.	G.4, G.5	A record of practice is meaningful only if it shows what the child drew (SRC-001; SRC-005, Moustafa, 6:03).
ULC-006	The project gathers no information from or about children and runs no study with children.	CON-005, CON-006	Directed by the supervisor (SRC-004); a study with children would need ethics approval that takes months (SRC-005, P2, 22:56).

28 Traceability

Insert your content here.

References

[1] Suzanne Robertson and James Robertson. Volere requirements specification template, 2012. Atlantic Systems Guild Limited.

29 Appendix — Requirement Sources

This appendix is the requirement sources register. Each source has a fixed identifier (SRC-001 to SRC-009) that is never renumbered. A requirement names the sources it comes from in its rationale or source field. A statement from the specialist interview is cited by its source identifier, participant code and transcript timestamp, for example "(SRC-005, P1, 28:37)"; the statement identifiers in Section 29.1 (for example SRC-005.01) can be cited instead. The interview is expert-reported evidence only. The three participants are de-identified domain specialists (P1, P2 and P3). No child took part in the interview or in any other source listed here.

Table 19 lists the project documents and meeting sources, and Table 20 lists the interview, questionnaire and peer-review sources. Requirements identify their supporting sources through stable source identifiers. Each source and statement has a LaTeX label (for example `src:005.01`), so a requirement can link to it and the link works in the PDF.

The sources represent users in different ways. Children are the main beneficiaries, but by the supervisor's direction (SRC-004) no requirement is gathered from children and no information about them is collected. Their needs enter the requirements only through adults who know them. The domain specialists' interview and questionnaire (SRC-005, SRC-009) are the outside evidence for these requirements. Their advice established adult control of lesson selection and optional stage changes, stylus as the main input, finger input as a stretch goal, and camera tracing as a stretch goal. Parents, educators and occupational therapists have not been consulted yet. The specialists advised doing so (SRC-005.03, SRC-005.06), and until then the requirements that depend on their views rest on the expert statements below and are open to change.

Table 19: Requirement sources: project documents and meetings

ID	Type	Date	Reference	Summary	Limitations
SRC-001	Project document	–	`docs/Problem StatementAnd Goals/ Problem Statement.tex`	Problem, stakeholders, goals and stretch goals.	Written by the team; some assumptions changed after SRC-005.
SRC-002	Project document	–	`docs/ Development Plan/ Development Plan.tex`	Team process, tools and schedule.	Written by the team; plans, not user needs.
SRC-003	TA meeting	Sep 23, 2026	Issue #10	Website instead of an installed app; teachers and students as primary stakeholders; specific, measurable metrics.	Short minutes; not a user source.
SRC-004	Supervisor meeting	Sep 23, 2026	Issue #21	Collect no information about children; do not gather requirements by asking children; ethics approval for studies; start with a web app; traceability.	Advice from the client, not observed user data.
SRC-008	Supervisor meeting	Oct 14, 2026	Issue #107	Let the supporting adult choose child-dependent thresholds within a team-set minimum and maximum, with example drawings of each end; default to the middle of the range.	Advice from the client, not observed user data.

Table 20: Requirement sources: interview and reviews

ID	Type	Date	Reference	Summary	Limitations
SRC-005	Specialist interview	Oct 1, 2026	Appendix 30	Three de-identified specialists on progression, input method, child-friendly design, feedback, accessibility and ethics.	Three experts, one session; expert opinion, not child observation; the experts advised confirming details with occupational therapists and educators.
SRC-006	TA meeting	Oct 7, 2026	Issue #78	State machine for the formal specification; define jargon; named TBD constants; specialists as de-identified secondary stakeholders.	Course guidance, not user needs.
SRC-007	Peer review	–	Issues #69–#75	Team 13 review of SRC-001 and SRC-002.	Reviewers' opinions; not all suggestions are adopted.
SRC-009	Specialist questionnaire	Oct 2–4, 2026	Appendix 31	Written answers on the creative space and refinement, motivation and frustration, letters, and concerns.	Two of three specialists; anonymous, so answers cannot be matched to P1–P3; expert opinion, not child observation.

29.1 Key Interview Statements

This subsection lists the interview statements most relevant to the requirements. Each is a short paraphrase, not a quote; the exact words are in

Appendix 30 at the given timestamp.

SRC-005.01 (P1, 28:37; P2, 29:18; 30:20–31:02)
Deciding automatically when a child is ready for the next level would need assessment and is too complex for the time frame. An adult who knows the child (a parent or educator) should choose the level. The team agreed to leave progression to the supervising adult.

SRC-005.02 (P2, 29:18)
In a past project, a teacher personalized each child's activities through a separate adult interface that also showed progress.

SRC-005.03 (P2, 31:42)
Which progress measures help an adult should be decided with occupational therapists or educational specialists, using milestones from the literature if they exist.

SRC-005.04 (P3, 24:34; P3, 25:36; P1, 26:46)
Holding a pen affects how children draw and the usability of the app. Finger input is the easiest start; a larger stylus and then a smaller one could be later difficulty levels if fine motor practice is a goal.

SRC-005.05 (P1, 34:13)
The app should be child-friendly: a child who cannot read yet should be able to enter a lesson, practise, restart and move back and forth using images and icons rather than text, without constant adult help. Adults handle setup.

SRC-005.06 (P1, 36:26)
Positive feedback wording and colours should be designed with feedback from educators and occupational therapists, built up bit by bit.

SRC-005.07 (P2, 37:39)
Audio feedback is a good idea; offering more than one modality is encouraged in inclusive design, with the user able to turn the audio off. Design it carefully with therapists' feedback, and consider adults with ADHD or autism as reviewers when children cannot take part.

SRC-005.08 (P2, 39:26)
Start with the Web Content Accessibility Guidelines (WCAG), then apply more specialized design heuristics for autism from the literature.

SRC-005.09 (P2, 19:05; P1, 20:27; P2, 20:47)
From a universal design view, the skills apply to all children, including those with motor or sight challenges; designing for disability serves many groups.

SRC-005.10 (P1, 13:42; P1, 15:12; P1, 16:13; P2, 16:45)
Camera finger tracing in the air practises gross rather than fine motor skills. It is useful but should come after the core touchscreen tracing when time is limited.

SRC-005.11 (P2, 22:56; P1, 24:03; P1, 33:00)
Testing with children needs ethics approval (about three to four months where P2 works). Meeting experts for feedback without involving children does not.

SRC-005.12 (41:51–42:48)
All three participants agreed to the use of the transcript for eliciting requirements.

29.2 Key Questionnaire Statements

This subsection lists the questionnaire statements most relevant to the requirements. Each is a short paraphrase; the full answers are in Appendix 31, labelled by respondent (R1, R2) and question (Q1–Q4).

SRC-009.01 (R1, Q1; R2, Q1)
The free drawing space is valuable. Refinement should stay subtle and support the child's own drawing rather than replace or correct it, so the child keeps ownership. Both respondents suggested that refinement be optional or adjustable.

SRC-009.02 (R1, Q2)
Activities motivate when they are achievable, predictable and visually engaging, with immediate positive feedback and some choice. They frustrate when they are too demanding, repeatedly highlight mistakes, need very precise movements or change the child's work unexpectedly. Squigglr should recognize effort and progress, not only accuracy.

SRC-009.03 (R2, Q2)
Requiring a pen or stylus rather than a finger may make drawing harder; finger drawing may be simpler.

SRC-009.04 (R1, Q3; R2, Q3)

Letters are a useful later stage, introduced gradually after shapes, with some flexibility in how precisely a letter must be traced. R1 suggested Arabic as well as English letters; R2 suggested letters and numbers, then tracing letters to form a word.

SRC-009.05 (R2, Q4)

The interface should be appealing and child-friendly while avoiding excessive visual or auditory stimuli.

SRC-009.06 (R1, Q4)

Involve children, parents, educators and occupational therapists in design and evaluation, and look beyond accuracy at engagement, frustration, independence, confidence and transfer to drawing without the app.

SRC-009.07 (R1, Q4)

Squigglr could serve a broader population, such as other neurodevelopmental conditions and rehabilitation; avoid restricting it to one condition too early.

30 Appendix — Specialist Interview Transcript

This appendix contains the transcript of a 43-minute remote group interview held on October 1, 2026, with three domain specialists. The specialists develop software for children with autism, ADHD and other learning needs, and hold doctorates in human-computer interaction and related engineering fields. The interview was conducted by Rayan Nasrallah and Moustafa Moustafa: a 10-minute presentation of the proposed application, followed by a semi-structured discussion. All three participants consented to the use of the transcript for eliciting requirements (see 41:51–42:48 below).

The transcript is de-identified at the participants' request. They are referred to as P1, P2 and P3. Their names, institutions, locations and other identifying details are replaced with bracketed placeholders. A personal health detail is removed. The automatic transcription was corrected where it misheard words, and filler words, false starts and short acknowledgements ("mm", "yeah") were removed. Each turn keeps the timestamp from the original recording (minutes:seconds), which is how requirement sources in this document cite it (for example, P1, 28:37).

Rayan (0:03)

First, I want to introduce myself and also Moustafa. I'm a student at McMaster University. I'm in my final year and I'm studying software engineering with Co-op. So it's an engineering program where we also take a year or so to do an internship program at a company here in Canada. So I'm on my final academic year, and we reached out to you guys because we're doing a project for our final year. It's called the Capstone. And we'll just introduce the project very briefly, and then we want to hear from you guys mostly and get your expertise on the topics we'll be discussing. I'll let Moustafa introduce himself as well.

Moustafa (0:54)

Yes, very similar to Rayan. Also, my name is Moustafa, 4th year software engineering as well. We're in the exact same program. Both kind of had the exact same story. We lived in [location redacted] our whole lives, or for Rayan, it's half of his life, the second half. And then we came to Canada for university. We've been here for five years now. Yeah, Rayan can start.

Rayan (1:16)
Yeah, I'll start sharing my screen. Can you guys see it?

P1 (1:26)
Yes. So both of you are also like final year students in university. Okay.

Moustafa (1:32)
Yeah, we're both on our 4th academic year, but 5th year with the 16 months we took in the middle for the co-op.

P1 (1:39)
Okay, I see.

Rayan (1:41)
Yeah. So last year, we both went for our internship for 16 months, like between our third year and our 4th academic year. So right now we're on our fifth year of university, but final, final year hopefully. Inshallah.

Rayan (2:04)
So I'll just get started. Today, we're going to introduce Squigglr, the application that we're building for our final year engineering project. And over the next 10 minutes or so, we're going to walk you through what we're planning and what we envision for this project moving forward. Let's start with who this app is for. So Squigglr is meant for a pretty broad group of children. That includes kindergarten and primary school kids who are still building their foundational drawing skills and children with autism, ADHD, or other support needs. And one thing we want to be careful about is that we don't want to assume two children with the same diagnosis need the same kind of support. Every child has their own abilities and their own preferences. Furthermore, around each child, you've got the adults who support them, like parents, guardians, and caregivers, educators and learning support staff, and school administrators. We want Squigglr to help them too. Before we jump into the concept of our application, I want to go through some of the research that shaped our thinking. First, when we say visual motor integration, what we mean is coordinating what you see with how you move. For example, seeing a square and then guiding your hand to draw it. The first paper is Daly, Kelley, and Krauss. They looked at kindergarteners and found that kids with stronger visual motor integration tended to be much better at copying letters legibly. So basically, these skills are closely tied to things like

handwriting. It's a correlation though, so it doesn't prove that practicing one improves the other. The second is Taverna and colleagues. They ran a 10-week program with kindergarten and 1st grade children built around short games like completing paths and tracks. And every activity had easy, medium, and hard levels. After the program, the kids' visual motor integration scores went up. There wasn't a control group, so we can't say for sure that the program caused it, but it's a good example of structured practice with graded difficulty, which is pretty much what we're trying to go for. So putting these together, they support the idea behind structured, graded drawing practice, but they don't show that Squigglr itself will get those results. That still has to be evaluated. The next two papers are about why we're paying attention to motor-related needs in children we're designing for. The 3rd paper compared handwriting between autistic children and children without autism. The autistic group had more trouble specifically with forming letters. So basically getting the shapes of the letters right. And within that group, their motor skills predicted how well they wrote. Then the final paper did a systematic review on children with ADHD. They found that fine and gross motor difficulties come up a lot, but the severity ranges from mild to severe. So for us, the big takeaway is really the variation. Not every autistic child or child with ADHD has motor difficulties. And the ones who do are all different. So instead of building one path per diagnosis, we want guidance and difficulty that can be adjusted. And we want kids to be able to practice as much as they need at their own pace. All right, so let's take the first look at the product. Squigglr has three parts connected to each other. First, there are guided activities. That's where kids practice lines, basic shapes, and simple drawings, and they gradually move from tracing to drawing without any guides. Then there's the creative space, where they can just draw freely and use those skills they've built outside of the structured exercises. And lastly, there's practice history, which keeps a record of attempts so parents and educators can actually see what's being practiced. And across all three, you've got adjustable support, history that's saved between sessions, and everything runs in the browser. Now, Moustafa will go through each of these, starting with how an activity progresses.

Moustafa (6:03)

Yes, thank you, Rayan. So we're going to be using the activity of learning

how to draw a square as our example here. First, the child would see the demonstration, so where to start and which way to actually go to draw it. Then they trace over a visible outline. After that, we can start taking guidance away step by step. So maybe they only get the corners or part of the outline. And finally, they would draw the square completely on their own with no guides at all. Kids can repeat any of these stages as many times as they want, and the support can be turned up or down. Now, one thing we haven't done yet is decide exactly when a child moves from one stage to the next. So we're not setting any accuracy thresholds or a required number of attempts. We think that's something that should be informed by specialists, not just [inaudible]. So on the next slide, this is a concept of a tracing activity in progress. So in this example, you can see the child has traced three sides of the square, but they missed the right side, for example. Instead of failing the attempt or just finishing it for them, the app highlights the part that's unfinished and gives a short prompt so the child can go back and complete it in the same attempt. There are two main principles here. First, we keep the child's actual drawing. We never swap in a perfect square and count that as them doing it on their own. And the support can be adjusted too. For example, how much of the outline is shown, or if it's just the corners, as mentioned on the slide before. Second, the feedback is based on predefined observations that we are manually deciding. We're not proposing an AI model that gives personalized advice on each drawing. It would just be basic math and algorithms to figure out what's wrong with the shape they drew and comment on it if needed. However, the exact implementation details are still a work in progress. Now, so alongside the lessons, kids also get a free creative space where they can draw whatever they want. In this example on the left, a child has drawn something that looks like a bird. The idea is that the app would take that and tidy it up just a little bit. So basically the lines would get a bit straighter, the curves get a bit rounder, but it's still the child's drawing. We're not swapping it out for a perfect clip art bird. And if the model can tell what it is, like here it's clearly a bird, it can use that to guide the cleanup. But even if it can't, it will still just clean up the lines and curves a little bit. The way we're planning to do it is we send the drawing coordinates to a lightweight AI model, and then it sends it back slightly adjusted, like it sends back the same thing with slightly adjusted coordinates at the same scale, and we rebuild the drawing inside the application itself. That way, everything stays editable

so the child can still move things, resize them, and change the colour or erase them, and they can always restore their original drawing, so they stay in control of their own work. This only happens in the creative space, though. In the structured activities that we talked about in the previous slides, we always keep the child's real drawing. The main goal for this creative drawing area is for it to be fun and to give kids a chance to use their skills that they learned outside of the exercises. Now here is the concept of what a parent or educator might see on our app. For every structured attempt or exercise, the record includes the activity and its guidance level, the child's drawing, any difficulties they faced, and the final outcome. So for example, in attempt one, the child missed 2 sections at first. But when they went back and completed them, the record keeps both of those. Attempt 2 was the same outline guidance, nothing was missed, and then a later attempt, so attempt 3 near the bottom, it said that was completed with fewer guides and it was fine. Adults can look at which activities were practiced, attempts over time, what guidance was used, and the difficulties that kept coming up or got resolved, and the actual drawing of the child. And there are some things we're deliberately leaving out. So no scores, no comparisons with the general population, no diagnostic labels, and no recommendations for special classes or treatment. We also don't want to suggest that fewer attempts automatically means better visual motor integration. The whole point is to just make practice easier to understand and to support better conversations about what a child needs. So now moving on to how we're thinking of building it, Squigglr is planned to be a web application. Basically, a child practices and draws on a compatible device that supports drawing, like an iPad or a tablet. Their attempts get saved to an account, and then adults can review that history later from any device. Part of the reason we went with the browser is that installing separate apps on school-managed devices can be pretty difficult, according to our supervisor who has some experience working with schools in Canada. We'll be targeting supported browsers and compatible devices, but we're not promising offline access or that it will work on every device just yet. Okay, so just to sum up the core idea, we have structured practice with lines, shapes, and simple drawings, progression from tracing to drawing without guides, guidance and difficulty suited to each child, free drawing with automatic refinement, accurate records of attempts and difficulties, and a clear practice history for parents and educators. There are also two extra features that

are not part of the main requirements, but we wanted to get your feedback on them as well. The first is camera-based finger tracing, where the shapes and guides show up over a live camera view, and the child traces them with their finger in the air. So instead of having to draw on the screen, they would just do it like this in the air, and it would work the exact same way. The second is taking the same approach of the guided practice we talked about for the shapes and the drawing, but applying it to letters as well. So our main vision is a welcoming place for children to practice drawing at their own pace, with a clear picture of that practice for adults who support them. So that's the whole app idea in summary. Thank you so much for your time. And before we get into our questions for you, please let us know if you have any questions or if anything we chose so far wasn't clear.

P2 (12:07)
Thank you very much.

P1 (12:07)
So thank you very much. Actually, I have a couple of questions. It's a very nice application. One of my questions, actually I have three questions. The first one would be, did you guys co-design this app with educators or therapists or people, like basically the end users who would ultimately utilize this application?

Rayan (12:27)
Moustafa, should I answer?

Moustafa (12:29)
Yeah, go, yeah, for sure.

Rayan (12:31)
So basically, like, we just started this project. We haven't built anything. We haven't even come up with the requirements. So our first kind of thing that we wanted to do is reach out to specialists like you guys that have expertise building apps like this. So the whole point of the engineering project is like we actually build a detailed requirements document over the next month or so. And then we start with the implementation towards December and hopefully finish it towards February and then do a usability test before the end of the second semester, before the end of April. So

we're still building the application, like we're still getting the requirements and everything. And the problem with working with kids, I'm sure you guys know, is there's like ethics involved in everything. So our strategy for eliciting requirements was more to go and ask people like yourselves and also like parents and caregivers and so on. So we haven't worked with kids or anything or educators yet, but like this is our first kind of step, if that makes sense.

P1 (12:42)

Okay.

Moustafa (13:41)

Yeah, and also just to add a small note, we also have our supervising professor for the Capstone Project, who also has some experience in this field. So he's partnered with a couple of organizations in Canada, and he has extensive experience with working with kids with autism and stuff like that. So he's going to be able to help us out as well.

P1 (13:42)

Okay. Okay, sounds great. I don't know why I thought that maybe this app was already being developed. The second question I have is a question that you actually proposed in the last slide about the camera, whether they should trace it using the camera. I mean, the camera-based finger tracing, exactly. This idea is nice, but I would say from my viewpoint, I think it might target a very different task. So whenever they're doing it in the air, that might be more of like a gross motor skill. And then whenever they're drawing on an iPad, that's more on the fine motor skills. So this is something that you could probably check with, you know, either therapists or educators and things. But this is my feedback on that. And you could integrate both. I mean, the camera-based finger tracing is more kind of like an augmented reality component that you would be including. And it's very interesting, especially with children who are more neurodiverse. But you just want to focus; they might be targeting different aspects.

Moustafa (14:55)

Okay. Okay, makes sense. But do you think it might be a good idea in general though to have it? Like, do you think there's actually something helpful about this feature if we do end up implementing it?

P1 (15:12)

I do, because a lot of times, even just looking at child development, before they start the very fine tracing, they usually have big markers, big things, and they're kind of tracing it very largely. So this would replicate that. And that kind of works more on the visual motor, even if it's not fine motor, it's more on the gross motor skills. And it helps them also in the visualization. So if they can visualize tracing a square, it might make it easier for them to, you know, trace it when it comes on an iPad and utilize their fine motor skills. So I think it's definitely an important feature to incorporate.

Rayan (15:48)

For now, though, because we might have a limited scope, like we have a certain amount of time to implement this app, do you think our focus with the shapes and the tracing and teaching them to draw is a good starting point and that we should focus on that before implementing this camera feature? Or do you think it should be part of this application from the get-go?

P1 (16:13)

I mean, if you're limited in time, I would say just focus on the first part, and then you could progress later on to extend it to the camera-based finger tracing.

Rayan (16:23)

Sounds good. Thank you for the feedback.

P1 (16:28)

I have a couple more, like some feedback, but I'll give maybe [P2] an opportunity if she has any questions and then I can continue.

P2 (16:40)

Thank you. It's really very nice.

P1 (16:43)

Your voice is very far, [P2].

P2 (16:45)

Can you hear me now? Can you hear me well? So it's really very nice work, both. It's just very inspiring what you're doing. So I agree with [P1]

in terms of focusing and leaving the camera and the tracking for maybe something to extend beyond this project. That's one thing. A second aspect, actually, while you were introducing it, I was thinking about my own self when I was very young in elementary school. I've always had difficulty in drawing a line. So when I saw this project, it's very nice. I think it's a very relevant problem. That's first. And my question to you: are you targeting specific groups of children with learning needs, or is it kind of broader? I would like to hear from you, who is your sample population?

Rayan (17:45)

So the idea started for kids with special needs. So kids with autism and ADHD, because the whole thing is each case differs. But from what I've seen personally, even as kids tend to get older, they still struggle with these basic skills that really young kids like toddlers and kids in kindergarten learn to develop early on in their lives, right? So our idea is we're mainly focusing on kids with autism and ADHD, but we also want to extend it to very young children, right? Like this is a skill that, like you said, you remembered when you were young. A lot of kids don't practice this when they're young. It's not integrated into learning. And I feel like it is a very necessary skill. So not just kids with special needs that are a little bit older, but also very young kids as well. We think they can use this application. And one big component of our design, which we wanted to ask about later, was how we can let the guidance and the support be controlled by the caregiver or the parent that's there supporting the activity. Or even if we should have a parent or a caregiver supporting the activity in the first place. So that's where it goes on to.

P2 (19:05)

Thinking about it from an inclusive, universal design perspective, this could be relevant to, yes, I agree, autism and ADHD, but then also children with gross motor and fine motor challenges, and also kids with sight challenges. [Personal account of a childhood vision condition removed.] So I believe this could be a training tool for any child in need of it, regardless of their situation, because there are many conditions that could affect our visual perception, and also affect even our handwriting. So just like when I saw this, I felt like...

Rayan (20:13)
Yeah. That's the plan, hopefully.

Moustafa (20:14)
So similar to what Rayan said, you don't think we should only limit it to children with special needs or autism or ADHD or anything like that? It should be general.

P2 (20:14)
Yeah, because...

P1 (20:27)
Yeah, I agree. I don't think you should limit it, because the skills that you're trying to teach now are applicable to all children, right? But I guess depending on their disability, they might achieve it at different stages in life. But if you go to nurseries that are looking at children at two and three years old, this is what they're focused on.

P2 (20:47)
Exactly. That's true. Actually, there is a saying in the world of accessibility: if you design for people with disability, you are actually designing for all. So whatever serves a certain population could be of use for many other populations. Then, thinking even of elderly people, people who get a stroke at some point in life, this is also very effective. I would see it in a rehabilitation centre, for example.

Moustafa (21:18)
Yeah, I didn't think of it, yeah, that's true.

P2 (21:25)
So the use cases could be endless, really. And I think that's also, if it's a startup, a selling point as well, because it could be put in a rehabilitation centre, it could be put in a kindergarten, it could be put in a school or at home.

Rayan (21:45)
Inshallah.

Moustafa (21:47)
Something happened to your mic. Yeah, yeah. I think it disconnected.

P2 (21:57)
We can see you, but we can't hear you.

Rayan (22:00)
Hello!

Moustafa (22:03)
Talk again.

Rayan (22:04)
Okay. Can you guys hear me?

Moustafa (22:06)
Yeah, that was it.

Rayan (22:07)
Yeah, yeah, I forgot what I was saying. Yeah, we wanted to ask actually if at some point we are able to build this app, would you guys, because you guys actually do research, would you guys see yourselves kind of looking into this area, or using this app, or doing further research? Like, if we just start with the idea, like a proof of concept, this app could serve as a proof of concept. Would you guys be able to take it beyond that in the future? And hopefully maybe in February, be able to actually get children to use it and do some observation, not necessarily a trial or research, but just observation activities, because it's difficult for us to do it here in Canada. That's why I wanted to ask.

P2 (22:56)
Yeah, but actually, it's equally difficult [here]. You need to get the ethical approval and the process is quite... Yeah, I think the laws have been tightening over the last 10 years, I would say, especially in this part of the world and in research. But it's easy for us to obtain approvals because we have experience, and an approval would take about three to four months really to obtain. But then actually, as you also speak, I am aware of a group at [another local university] that is looking at handwriting and dyslexia and other learning [needs]. So there is also a very active group there that looks at learning needs and technology, the use of image processing in particular. I can share their contact with you if you're also interested to meet more expertise in the field. And I also

wanted to say something else. Well, I forgot, but these are the two points from my side, but this is really nice, very nice work.

Moustafa (22:59)
Yeah, I think it's difficult everywhere.

Rayan (24:03)
Thank you.

Moustafa (24:03)
Thank you so much.

P1 (24:03)
Yeah, absolutely. I think it's great work. I think it's much needed. And I mean, we would be happy to kind of help guide you. I think there is some merit in trying to, you know, pilot or test this application. Of course, you would need an IRB, but maybe you could do this where you're at, through your university, right? And we could support you through the process and kind of help guide you. So I think it's a great application.

Rayan (24:26)
Yeah, that would be perfect. Yeah. Thank you guys. Appreciate it.

P2 (24:31)
Thank you.

Rayan (24:32)
[P3], you.

P3 (24:34)
Yes, I just want to ask one thing. Honestly, I'm curious, I'm not sure if I missed it or not while you were presenting. Will you ask the children to hold the pen in order to do the line, like the drawing, or will it be their finger? Like, will it be touch, or are you thinking of a pen?

Rayan (24:56)
I think it can be either. We haven't really got into that decision. Our initial thought was a pen. Like, I have this iPad here. So like this, for example. And then maybe as well, there's attachments you can buy for these that make the grip wider. Because I know some kids have trouble holding it like this. So I know there's grips that you can buy that make

it large. So it's very flexible. I feel like it should support tracing with the finger and pen. But what do you think?

Moustafa (25:03)
Thinking about it now, a pen is probably not a good idea, right?

P3 (25:11)
Exactly, yeah.

P3 (25:36)
And that's why I ask, because as [P1] was saying and [P2] also was saying, regarding the fine motor skills of the children, like how they will hold the pen? Will they hold it from the top, or some will hold it like this? You know, I don't have a pen near me, but they will hold it like this. So this might affect how they will draw, which might affect maybe the usability of the app itself. So if your goal is not about the fine motor or something, we could remove the pen. If your goal is more about the drawing itself, only the drawing idea, I guess you can use the finger thing. Not sure if [P2] and [P1] agree with me on this idea, but just to make your app more reliable in real life for the children.

P1 (26:35)
So you mean to draw with their finger on the application?

P3 (26:40)
Yeah, it will be like a touch. They will draw as Moustafa was saying in space using the camera. So it will be the same but on the tablet or iPad itself, rather than holding a pen and having more challenges.

P1 (26:46)
In the air, yeah. That may be a start. I see that. That's a good option for a start. But again, if the idea is to kind of improve the fine motor skills, they have to practice with being able to hold something. So usually they start with something large, because it's easier for their hands, and then they have to go to the smaller ones, like pencils. That's more difficult, so that actually takes some training for them to do. So you could embed difficulty levels, you know, where they start with their finger, which is the easiest, and then with a larger kind of marker or pencil, and then something smaller that's more reflective of what they would use in schools.

Rayan (27:31)

So on that point, we had a question about how we should adjust, because in our thought process, we don't want the app to judge when a child is ready to move from one stage to the next. So the example we gave throughout the slides was the square. But of course, a square is a very simple shape. And then they could move to triangles and then stars and more complex objects that they can trace and then draw by themselves, right? And we'd have an accuracy threshold. But our concern was, should we dictate when the child is ready to move from a square to a triangle or a square to a star? Or is that something we should leave fully to the supervising adult during the activity? And what determines when a child is ready to move on? Because now you introduce an extra degree of complexity, which is holding an object in and of itself; decreasing the size of that is also progression, but that's not something we can necessarily track.

P1 (28:37)

So if you're going to look at that, then you're going to open up a door of assessments, right? Because there needs to be assessment. And I think that, given your time frame, it might be too complicated at this stage. I would keep it user interactive, so the user can decide whether they're going to go from, I don't know, level one, level 2, level 3, right? And then ultimately who decides what they're going to do would be either their parent or the educator or somebody who knows where their level is at. And then they could tell them, you know, you need to practice level one until you're proficient in it, and then they can ask them to move. But if you want the app to do it automatically, that's going to be more advanced.

P2 (29:18)

Actually, just something we had in one of our projects: we had an app for children to learn vocabulary, and we co-designed it with a centre [location redacted] with speech therapists in the centre, and we allowed the teacher to kind of personalize the journey for each child separately, and that's not really automated. The teacher would have a back-end kind of interface to do that. But as [P1] said, that actually will bring you to the other side of the story, which is assessment. But then you might want to think about just allowing the teacher or the caregiver to kind of change the activity

and giving them a back-end view of the progress. But then, thinking also, if you do that, then one would ask, oh, what kind of assessment are you applying here?

Rayan (30:20)

Yeah. So I think maybe the best thing is to do what you guys recommend, which is to kind of leave it to the caregiver or the supervising adult, and we shouldn't worry about assessment right now. It shouldn't be in the scope given the time frame. Maybe something in the future, Moustafa, I don't know if you agree.

Moustafa (30:38)

Yeah, I feel like it's very good to have, but maybe not in these short months. It would be nice, but I don't think we have time to do it right now. So right now, as you guys recommended, it would probably be the adults who would say, okay, this is fine for now, move on to level 2 or level 3, instead of us having to figure out how to map when this child is ready for level 2 or level 3. We can figure that out later.

Rayan (30:52)

Hundred percent, yeah.

Rayan (31:10)

Speaking of that, in the history that we showed, what statistics or what values or what information would actually be useful to the parent or caregiver, so they can say, okay, this progress history is actually benefiting my judgment of whether the child is ready to progress? For example, should we have a percentage, like, okay, the accuracy threshold is above 80%, we passed it, but we're still showing you that this is the accuracy of the kid's drawing?

P2 (31:42)

That's a very good question. Really, I think this is a question to ask the therapists or the teachers, if there are certain defined milestones in the literature that you could kind of have as your benchmark or reference there. And that's something to discuss really with, I think, either an OT, an occupational therapist, or an educational specialist.

Rayan (32:12)

Yeah. Do you guys know any occupational therapists or educational

specialists that we can speak to about something like that, that you could connect us with?

P2 (32:21)

I think the easier way to do that is reaching out to private centres [location redacted]. I think [P3] can share them with you step by step. They are very good, they are very approachable. You might want to introduce this, and they are very interested in technologies. So, maybe that.

P1 (32:42)

And [a local hospital] as well. They have a therapy program where they have occupational therapists as well.

Rayan (32:50)

We're scared that if we email them, we might scare them or something. They might start talking about ethics and stuff, but it's just a small project. It doesn't seem like it, I guess, when we reach out to them, but it's hard to kind of...

P1 (33:00)

Well, the ethics only comes in whenever you say that you want to test your product, you want to meet with the kids and stuff. But if you're asking just to meet with them and get their expertise and get their feedback, it should be okay as long as you're not really interacting with the children at this point. Just to get feedback through a meeting, like what we're doing now, it should be fine.

Rayan (33:26)

Okay. Sounds good. Moustafa, do you have any more questions you wanted to ask?

Moustafa (33:34)

Yeah, this was on a point that we were mentioning before. So with deciding when the child goes on to the next level and stuff like that, a similar question: how should this app be made? Should this app be built in a way that the child by himself can go in and kind of know how to play around and move from level one to level 2, or even go to the creative mode and draw? Or are we expecting that there's always a parent present? Because we've never worked with kids with disabilities, or not even kids with disabilities, just children in general, very young

people. We don't know how to give feedback, like, oh, good job, now try this. Or how do we tell them to draw a square? Or how do we tell them to do this? Or are we assuming that there's a, yeah?

P1 (34:13)

That's a very good question. Yeah, so to answer the first part of your question, I think you should design the application so that it's child-friendly. Because if you're going to design it in a way that the parent has to keep, you know, interfering, parents are going to quickly get annoyed, and, you know, if they give them the iPad to practice, they want them to practice without their constant interference. So then you have to think about how you design the application so that it's child-friendly, meaning that, you know, both the child and the adult can understand the interaction. So a child who's three years old maybe doesn't know how to read yet, but, you know, kids these days are very tech-savvy, so they know how to navigate applications and things. But if you look at the games that they play, it's usually not very text heavy or oriented, so they have images and icons that they navigate in order to go and play. So you would have to design it from the get-go in that kind of manner. Adults obviously will know how to navigate through and install and do the initial parts, but you know, you should also design it in a way so that children can navigate and be able to enter the lesson that they're on, be able to practice, to restart, to go back and forth. Because otherwise, if you make it too high level, where they require constant adult supervision, I think you're going to limit the usability of it.

Moustafa (35:31)

Okay, so it should be somewhat expected that the child can use it alone, play around with it.

P1 (35:36)

Yes, and then, yes.

P2 (35:36)

I think also, sorry, [P1]. It's just, I think also it's your decision on what you want the app to look like. Like, you know, there are many drawing apps in the App Store, right? For children, for mainstream, or maybe not even in mainstream school, it could be used in a centre. But it's your decision to say whether you want this app to kind of follow a process and

have, yes, a child as a user, but then the caregiver can give them the access. So I think these different workflows, you can put whatever you like, because there is no right and wrong, sort of. There's only the way you present it and the angle in which you're presenting it.

Moustafa (36:22)
I see, I see.

P1 (36:26)
No, no, there's no problem at all. I think the next step would be for you guys to meet the end users and the stakeholders, which would be the educators and occupational therapists. They can also give you more feedback. And they're the ones who will be able to guide you, because we're developing similar, not similar, but I mean, similar types of applications, one for occupational therapy. And even the way that we give them positive reinforcement, like if they get something correct, what we say and how we say it and the colours that we use. I mean, we go through the literature, of course, but we also get the feedback from, you know, the experts, because we're doing the design and the technology aspect, but they're the ones who deal with the patient side or the children's side on a day-to-day basis. So these kinds of design aspects, you can build bit by bit.

Moustafa (37:15)
I see. Okay, thank you so much. Rayan, you had something to add.

Rayan (37:18)
I just had a follow-up, yeah, so... The first thing is, what do you guys think about voice, like dictation, like voice support for the kids? Like saying, oh, you did a good job, with like some sound? Or do you guys generally see that that's not a good idea?

P2 (37:39)
No, actually, it is a very good idea. We had an app, the vocabulary learning one, that had a voice input as well. And multi-modality in general, like having two modalities to communicate, and then the user could choose; he could turn off the audio. So multi-modality is very much encouraged in the field of design, right? If you want to design an app... now if you look at the Apple products and the phones in particular, they

also have the haptic modality, so the phone can shake sometimes if there is a notification. So multi-modality is always encouraged, even in the realm of inclusive design and universal design. It's always encouraged, but it's just, as [P1] said, designing it carefully, getting the feedback from the therapists, and even, if you have the chance, getting the feedback from the users themselves. Even if you cannot reach the children as a sample or a group, you could reach an adult group with ADHD and autism. And if you want to expand it to different adult groups, then that's also useful, because then you based your decision on real experiences.

Moustafa (38:58)

It's a very good idea, okay.

Rayan (38:58)

Okay. And then I have a follow-up to that as well. In general, what design choices in apps tend to cause problems for kids with autism and ADHD? And also, what are best practices for these types of apps? Because you all have worked on HCI and these types of apps, making sure the interface is specific to their needs. So what general patterns do you guys recommend we should follow?

P2 (39:26)

So yeah, this is a very good question. I teach an elective course for the undergrads in the spring on ICT accessibility, so how to design accessible interfaces and apps, and the main set of guidelines is called the Web Content Accessibility Guidelines. We call it the WCAG. And it's very famous. It actually is adopted as law in many countries, including the US, and [location redacted] adopted it last year. So I think start with the WCAG and then move forward to more specialized guidelines too. Start with the WCAG and then look at design guidelines for autism in particular; we call them heuristics. So heuristics for autism, app design for autism. There is a lot in the literature; just write in Google Scholar "heuristics for app design for autism" and you'll find a lot. But I would suggest starting with the WCAG and then moving to the more detailed ones.

Rayan (40:39)

Okay, perfect. Thank you. Thank you.

Moustafa (40:39)
Your course would be very helpful for us right now, the accessibility course. I wish, I wish.

P2 (40:44)
Yeah, you're welcome to attend, welcome to come [location redacted].

Rayan (40:44)
Yeah. Yeah. Hopefully, that's the dream. Inshallah. Do you guys have any more comments about that? Like, [P1], [P3]? Do you guys have any advice or any patterns that you guys think of?

P1 (41:08)
I mean, from my end, no. I think what you're doing is really great and it's very timely, and we would be happy, maybe after a couple of weeks or a month, to meet again. And if you have questions, we can again give you our view and feedback, just to kind of help guide you through the process.

Rayan (41:26)
Yeah, that would be perfect. I mean, we have a lot more questions, but because we're running short on time, would it be okay if we sent you guys a form or something that you can answer on your own time, whenever you have time?

Moustafa (41:26)
That would be amazing, yeah. It would probably be just very, very short questions.

Rayan (41:44)
Like, they're very specific.

P2 (41:49)
Yes, of course.

P1 (41:51)
Yeah, that's fine.

Rayan (41:51)
Okay, perfect. Well, we'll create a form then, and we'll send that out to you guys, and it has some more questions that we didn't really get to ask.

And also, the whole purpose of this interview, like we said, was to kind of get your feedback and then elicit requirements from that. I guess now I'm realizing we have a lot more work to do. We have to talk to some more people before we can actually elicit the requirements properly. But I was going to ask you guys if you could sign a form, a consent form, that we can use this transcript to basically do that work. We'll send it to you guys as well, with the questions, at the same time, by email, if that's fine.

P2 (42:37)
Yeah, of course.

P1 (42:38)
No, it's fine.

Moustafa (42:39)
Yeah, it's basically just saying that we can use the transcript of what we talked about here to elicit the requirements, and that we had this meeting, basically. That's all.

P3 (42:40)
Sure, no problem.

P1 (42:47)
Sure.

Rayan (42:48)
Okay, perfect. Thank you guys so much.

Moustafa (42:49)
Thank you so much. Also, does anyone have any other questions before we end?

P1 (42:52)
No problem.

P3 (42:56)
Thank you so much. That's amazing work.

P1 (42:58)
No problem. Best of luck [inaudible].

Rayan (43:00)
Well, we'll be reaching out to you guys again, hopefully in the near future. Inshallah.

Moustafa (43:00)
Thank you guys so much. This was very helpful. Inshallah.

P3 (43:06)
Inshallah, Inshallah.

Moustafa (43:08)
Thank you so much. Thank you so much.

P1 (43:09)
Thank you.

Rayan (43:10)
Thank you guys for your time. Bye-bye.

P3 (43:11)
Thank you, bye.

P1 (43:14)
Bye.

31 Appendix — Specialist Questionnaire Responses

This appendix contains the written responses to a follow-up questionnaire sent to the three domain specialists who took part in the interview (Appendix 30). Two of the three responded. The form was anonymous, so the responses are labelled R1 and R2 and cannot be matched to P1–P3. The responses are reproduced as submitted. The form export cuts off the end of the first and fourth questions, shown here as "…".

Q1

Regarding the free drawing space we showed, where a child can draw anything and AI gently tidies it up; do you think this is helpful for children, or unnecessary? Its meant to be a fun area where…

R1 I think the free drawing space could be very valuable, particularly because it gives children an opportunity to explore and practice without the feeling that there is one "correct" answer. However, I would be cautious about how much the AI modifies the child's drawing. The aim should be to support the child's own effort rather than replace or correct it. I would prefer the AI to provide subtle assistance, suggestions, or optional touch-ups that the child can accept or reject. Preserving the child's sense of ownership and agency is very important. It may also be useful to allow the level of AI assistance to be adjusted depending on the child.

R2 I feel it is a very nice idea so that the children could draw anything and the AI could let's say make it better. However, sometimes the unclear lines are important for the imagination of the child. For example, a family picture with unorganized lines will look better (in the child's perspective) than a very clean drawing. Hence, I feel this should be also optional feature and not mendatory.

Q2

From your experience, what makes practice activities like this motivating for these children, and what makes them frustrating?

R1 Motivation often comes from activities that are achievable, predictable, visually engaging, and provide immediate positive feedback. Choice is also important—for example, allowing the child to choose what to draw, the theme, or the level of difficulty. Small signs of progress and encouragement can be motivating without making the activity overly competitive.

Frustration can arise when the system is too demanding, repeatedly highlights mistakes, requires very precise movements, or changes the child's work in ways they do not expect. It is particularly important that the AI does not unintentionally communicate that the child's drawing is "wrong." The system should accommodate different motor abilities and drawing styles and should recognize effort and progress, not simply accuracy.

R2 The drawing activity itself is likely to be a key source of motivation. However, the experience may become frustrating if the activity includes too many demanding or overwhelming elements. For example, requiring children to draw using a pen or stylus rather than their fingers could make the interaction more difficult, whereas finger-based drawing may provide a simpler and more intuitive experience.

Q3

Would extending Squigglr from shapes, to tracing and forming letters be useful?

R1 Yes, definitely. I think this could be a very useful extension, particularly if it is introduced gradually from simple shapes and strokes to more complex letter forms. There is a natural progression from basic pre-writing shapes to letter formation. However, I would recommend maintaining flexibility rather than expecting one very precise tracing pattern. It could also be interesting to explore Arabic as well as English letters, particularly in our context, as the motor patterns and directionality are different.

R2 Yes, I believe this could represent a natural second stage in the development of the application. The initial phase could focus on establishing the effectiveness and usability of free drawing and basic shape-tracing

activities in supporting children's engagement and learning. Once this concept has been validated, the application could be extended to include letters and numbers.

A more advanced mode could then introduce sequential tracing activities in which children follow individual letters to form a complete word. For example, the child could trace the letters "B-O-Y," after which the application displays an illustration of a boy. To further enhance engagement and reinforce the connection between the written word and its meaning, the illustration could also include a simple animation, such as the boy waving.

Q4

Now that we've talked through everything, what stands out to you most about Squigglr: what worries you, and what seems most promising? Anything we didn't ask about that you think we should consider...

R1 I would also strongly recommend involving children, parents, educators, occupational therapists, and other relevant professionals throughout the iterative design and evaluation process. Beyond measuring whether drawings become more accurate, it would be valuable to examine engagement, frustration, independence, confidence, and whether the skills developed through Squigglr transfer to drawing and writing without the system.

As I also mentioned during the meeting, I see potential for Squigglr to serve a much broader population. Its underlying concept could be relevant across a range of neurodevelopmental and neurodegenerative conditions, as well as rehabilitation contexts such as stroke and acquired brain injury, and for individuals with learning difficulties or motor challenges.

For this reason, I would suggest not restricting the development of the app too early to one specific condition or case study. Instead, I would focus on clearly defining the core capabilities and value that Squigglr can offer—such as supporting fine motor skills, visuomotor coordination, tracing, drawing, and handwriting—and then explore how these capabilities can be personalized and evaluated across different

user groups. This could significantly broaden both the impact and future potential of the application.

R2 The application itself is very promising. However, one concern I have is that the current design may not be sufficiently appealing or engaging for children. I would therefore recommend dedicating substantial attention to the user interface and overall visual design of the application.

The aim should be to create an interface that is engaging, enjoyable, and child-friendly, while at the same time maintaining a simple and comfortable interaction experience. In particular, it would be important to avoid excessive visual or auditory stimuli that could make the application overwhelming. Finding the right balance between an attractive and playful design and a calm, minimally stimulating environment could significantly improve the children's overall experience and willingness to engage with the application.

32 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

32.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

32.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

32.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

Individual Disclosures

Rayan: I used Anthropic's Claude Opus 5.5 at medium effort, accessed through Claude Code, while preparing my sections of this document. I developed the ideas and made the project decisions in my sections myself, and Moustafa and I conducted the specialist interview and follow-up questionnaire. A major use of AI was organizing our sources: I used the model to sort through the interview transcript, questionnaire answers, and meeting notes, and to extract the specific statements, with their timestamps, that supported the ideas I wanted to include. This would have been very difficult to do by hand given the length of the material. I also used it to turn my notes and decisions into draft wording, to improve sentence structure, clarity, and flow, and to help with LaTeX formatting. Through Claude Code, it also helped manage the repository (branches, commits, and merges) and helped me review my teammates' pull requests by checking their changes against our decisions and the rest of the document; I decided which review comments to post.

All ideas and decisions in my sections are my own. I reviewed every AI-assisted passage against my intended meaning and checked each cited statement against the original transcript, questionnaire, or meeting notes to confirm that it says what we claim it says.

33 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

   Rayan: What went well was that most of my sections could be built on what our stakeholders had actually told us, rather than on our own guesses. Moustafa and I interviewed three domain specialists on October 1, 2026, and the de-identified transcript is now an appendix to the SRS. Because of this, Sections 1, 2 and 5 and the requirement sources in Appendix A cite the specialists (P1 to P3) by timestamp, so a reader can check exactly where each statement came from. Our meetings with our supervisor, Dr. Anand, on September 23 and October 14, 2026, and with our TA on September 23 and October 7, 2026, gave us the same kind of traceable source for the constraints and business rules. It also helped that I split my part of the SRS into one issue and one pull request per section, which kept each review small and let each section be checked and merged on its own. The main thing I took from this is that recording sources as we go makes the requirements much easier to justify and to change later.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   Rayan: The main pain point was that decisions kept changing as we learned more, and each change affected several sections at once. In our interview on October 1, 2026, the specialists said that the supporting

adult, not the app, should decide when a child moves to an easier or harder stage, so we changed our progression goal from automatic to adult-controlled. In our meeting on October 14, 2026, Dr. Anand advised that thresholds such as the pass criteria and the number of failed attempts should be set by the adult for each child within a range the team defines, which changed several constraints, business rules and goals. The answers to the follow-up questionnaire that Moustafa and I sent the specialists, returned between October 2 and 4, 2026, then confirmed that letters belong in a later stage after shapes, so we kept them as a stretch goal and recorded the order of stages as a likely change. I resolved this by updating every affected section in the same pull request as each change, and by recording anything still unsettled in Open Issues (Section 18), so that nothing was left half-changed. A smaller problem was that several of my branches were open at once and each added a row to the revision history, so later pull requests conflicted on that table; I merged main into each branch and kept every row. I also found that some early drafts included details nobody had actually discussed, so I went back through each section and removed any claim I could not trace to a source or a team decision. Looking back, I learned that requirements are never really finished early in a project, and that keeping each change in one place and tracing it to its source is what makes them manageable.

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

   Rayan: In my sections, I wrote 24 requirements: 7 mandated constraints, 9 business rules and 8 safety requirements. Of these, 18 cite at least one conversation with our stakeholders or their proxies. Ten come from our interview with the specialists on October 1, 2026, including that adults choose the lesson and stage, that guided lesson records keep the child's actual strokes, and that Squigglr gives no diagnosis. Eleven come from our meetings with Dr. Anand on September 23 and October 14, 2026, including that Squigglr is a website, that no information is gathered from or about children, and that the adult sets each child's thresholds within a team-set range. Some cite both. The other six come from our own project documents and the course requirements, such as the deliverable deadlines. Since none of these stakeholders are the children themselves, nearly every requirement that affects how a

child uses Squigglr rests on what adults and specialists told us, which is why we plan to check them with the domain specialists again later.

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

   Rayan: Two courses help directly. The first is SFWRENG 4HC3, Human-Computer Interaction, which I am taking now and where our group's term project is Squigglr itself. The user research we do there, such as planning interviews with caregivers and educators and building personas, feeds straight into the stakeholders and requirements in this document. The second is SFWRENG 3RA3, Software Requirements, which I took in my third year. The requirements document we wrote for that course followed Meyer's template rather than Volere, but the concepts carry over directly. In 3RA3 we learned to separate hard goals, which need a clear, measurable and verifiable satisfaction criterion, from soft goals, which have to be broken down until each part can be measured. That is the same idea as the fit criterion Volere attaches to each requirement. We also learned to check every requirement against three qualities, that it is necessary, verifiable and attainable, and to write requirements with the EARS sentence patterns, such as "When an event occurs, the system shall. . . ", so that the wording stays precise. Being able to transfer those concepts helped me a lot with this deliverable in particular, because I already knew how to write goals and requirements that are precise and checkable and only had to adapt them to a new template.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

   Rayan: The most important knowledge the team needs is domain knowledge about the children Squigglr is for: how autistic children and children with ADHD develop visual-motor skills, how they use apps, and what helps or frustrates them. None of us has a background in this area, and it is not something we could safely guess, because a

wrong assumption could make the app harder or more upsetting for the children it is meant to help. That is why it was necessary to get input from people who have designed software for autistic children and children with ADHD, rather than relying only on our own ideas. Their experience is what turned our general idea into requirements we can actually justify.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

   Rayan: The two approaches we chose for acquiring this domain knowledge were interviews and questionnaires with domain specialists. Moustafa and I pursued both. We interviewed three specialists on October 1, 2026, because a live conversation let us ask follow-up questions and explore topics we had not expected, such as who should control a child's progression. We then sent them a written questionnaire, answered between October 2 and 4, 2026, to cover the topics the interview did not reach, such as refinement in the creative space and letter activities. The questionnaire also gave the specialists time to think through their answers. Using both worked well together: the interview showed us what we did not know, and the questionnaire filled in the gaps it left.